

**AI-Based Short-Term Generation Forecasting of Cascaded Hydroelectric Power
Plants**

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CERTIFICATE OF PANEL APPROVAL

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Abstract

The hydraulic coupling of the Agus 1–7 hydropower cascade in Mindanao makes day-ahead capacity nomination highly sensitive to hydrologic variability and operational uncertainty, often resulting in conservative estimates and frequent re-nominations. To address this challenge, this study develops and validates an Artificial Intelligence-based short-term generation forecasting framework using a Radial Basis Function Neural Network (RBFNN). Historical operational and environmental data—including plant generation, reservoir elevation, hydraulic flows, gate operations, rainfall, and unit-level availability—were preprocessed, standardized into a uniform hourly time series, and enhanced through feature engineering techniques incorporating temporal patterns, lagged variables, rolling statistics, and cascade-aware upstream dependencies. The forecasting model adopts a residual learning formulation, predicting generation changes relative to current output to improve continuity and stability. Model performance was evaluated using Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and coefficient of determination (R^2) under a chronological data split. Results show that the RBFNN achieved an average testing MAPE of 2.519%, with corresponding MAE of 1.355 MW, RMSE of 3.575 MW, and R^2 of 0.925, outperforming benchmark models including Random Forest and XGBoost across all Agus plants. The findings demonstrate that the proposed framework provides accurate, reliable, and operationally applicable forecasts, supporting improved capacity nomination, reducing re-nomination risks, and enhancing short-term planning in cascaded hydropower systems.

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CHAPTER 1

INTRODUCTION

1.1. Background of the Study

Hydropower remains a cornerstone of the Philippine energy portfolio, providing a significant share of total generation and critical support for grid reliability (Polestico, 2019). Central to the national strategy is the Agus–Pulangi Hydroelectric Power Plant complex in Mindanao, a major government-owned facility operated by the National Power Corporation’s (NPC) Mindanao Generation (MinGen) Group. The complex, with a total installed capacity of 1,001.1 MW, is sustained primarily by the hydraulically coupled Agus cascade along the Agus River, supplied by the vast storage volume of Lake Lanao (Polestico, 2019).

The Agus cascade—comprising Agus 1, 2, 4, 5, 6, and 7 plants—is a highly interdependent system, not a collection of stand-alone units, with a combined capacity of about 746.1 MW (Polestico, 2019). Water released by the upstream plants immediately dictates the inflow and hydraulic head available for all downstream plants as the river flows 36 km down to Iligan Bay (Polestico, 2019). Consequently, any operational decision, such as changes in gate opening or unit loading at one plant, directly and non-linearly influences the water available and the potential generation capacity of every plant below it. This cascading arrangement necessitates that operation, optimization, and power generation must be planned and modeled simultaneously for the entire system, making accurate short-term forecasting essential for scheduling and market participation (Polestico, 2019).

1.2. Statement of the Problem

Capacity nomination for the Agus hydropower cascade (Agus 1 to Agus 7) is frequently unstable. In practice, day-ahead nominations often need to be revised because the actual generation depends on a combination of hydrologic and operational factors such as forebay elevation, reservoir inflow and outflow, spillway and governor gate positions, riparian flow, and rainfall, which are not always measured consistently or fully reflected in

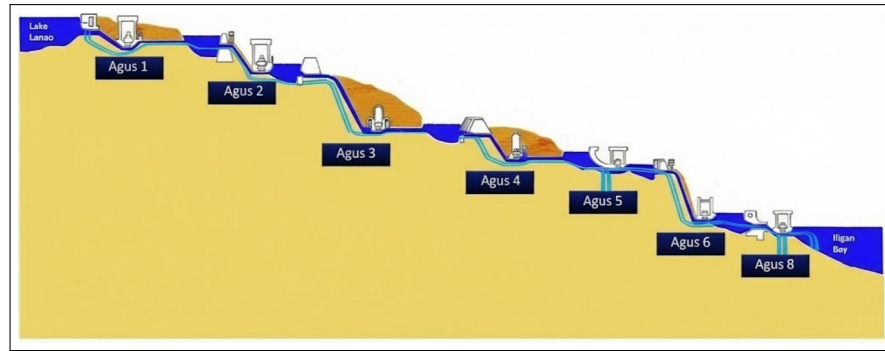


Figure 1.1. Schematic Representation of the Agus River Hydroelectric Power Plant (HEPP) Complex (National Power Corporation - Mindanao Generation & Mindanao State University - Iligan Institute of Technology, 2025).

existing planning tools. When these inputs are incomplete, outdated, or not systematically used, the current simulation models and rule-based practices struggle to represent the true behavior of the cascade, leading to conservative nominations, frequent re-nominations, and potential underutilization of available water and capacity.

1.3. Objectives of the Study

The main objective is to use an AI-based short-term (one-day-ahead) generation forecasting model for the cascaded Agus 1–7 hydropower plants using an RBFNN and to investigate how these methods improve the accuracy of hydropower generation forecasting driven by data on plant generation, forebay elevation, inflow and outflow, spillway and governor gate positions, riparian flow, and rainfall. Specifically, the study aims:

1. To design and develop an RBFNN-based short-term (one-day-ahead) generation forecasting model for the Agus 1–7 hydropower plants.
2. To evaluate the forecasting performance of the developed RBFNN model using statistical accuracy measurements using MAPE by comparing the results of the developed forecasting system with the actual data.
3. To compare the performance of the RBFNN model against benchmark methods to determine its improvement in prediction stability.

1.4. Significance of the Study

The hydraulic coupling of the Agus cascade renders day-ahead capacity nomination highly sensitive to hydrologic variability and operational uncertainty (Polestico, 2019). Accurate short-term generation forecasting is, therefore, an essential tool for supporting scheduling and water management (Polestico, 2019). The findings of this study will benefit the following sectors:

This study will primarily benefit the National Power Corporation - Mindanao Generation (NPC-MinGen) by providing a practical, operations-oriented forecasting model that mitigates the limitations of rule-based scheduling (Polestico, 2019). By providing more accurate one-day-ahead forecasts, this study supports NPC-MinGen in reducing the frequency of re-nominations and minimizing the risk of unnecessary spillage. Furthermore, in the context of the Wholesale Electricity Spot Market (WESM), the proposed RBFNN framework enables data-driven decision-making for generation bidding, allowing the utility to formulate informed day-ahead offers and maximize revenue potential (Polestico, 2019).

For the Mindanao Power Grid and System Operators, this research contributes to the overall stability and reliability of the region's energy supply (Polestico, 2019). By integrating upstream releases, travel time, and reservoir head dynamics into the forecast, the model provides system operators with a clearer picture of the expected hydro contribution for the next 24 hours. This improved visibility supports robust dispatch decisions, efficient reserve planning, and the maintenance of the region's energy balance (Polestico, 2019).

The academic community and future researchers will benefit from the advancement of theoretical and practical knowledge regarding Artificial Intelligence applications in renewable energy. This study addresses the specific gap in dedicated data-driven models for coupled generation forecasting in cascaded systems (Barzola-Monteses et al., 2019; Bernardes Jr et al., 2022). The methodology—specifically the rigorous preprocessing pipeline and the integration of multi-plant hydraulic features into an RBFNN architecture—serves as a reference for future researchers developing hybrid models for complex, non-linear hydrological systems (Polestico, 2019).

Finally, this work aligns with global policy and sustainability goals, specifically the United Nations Sustainable Development Goal (SDG) 7: Affordable and Clean Energy (World Bank, 2020). By optimizing the operation of the Agus cascade through better forecasting, this study promotes the efficient utilization of existing renewable resources (Zhou et al., 2020), thereby supporting the sustainable energy security of the region without the immediate need for new infrastructure construction (Lee et al., 2016).

1.5. Scope and Limitations

This study focuses on the Agus River hydropower cascade, specifically Agus 1 to Agus 7, as operated by NPC–MinGen. Other power plants within the Mindanao grid are not included in the modeling and analysis. The forecasting task is limited to short-term, one-day-ahead (24-hour) generation forecasting using historical operational and environmental data, including plant generation, forebay elevation, inflow and outflow, spillway and governor gate positions, riparian flow, and rainfall. No additional sensors or field measurements are introduced in this study.

The modeling approach utilizes a Radial Basis Function Neural Network (RBFNN) as the primary forecasting method, while other methods, if considered, serve only as simple benchmarks. The study does not develop a full physical or hydrodynamic model of the Agus–Lake Lanao system. Furthermore, evaluation and implementation are confined to offline analysis using historical data and statistical accuracy measures, particularly Mean Absolute Percentage Error (MAPE). Real-time system integration, detailed economic or market analysis, and Wholesale Electricity Spot Market (WESM) bidding simulations are beyond the scope of this thesis.

1.6. Theoretical Framework

The present methods used in hydropower generation forecasting each have their own strengths and weaknesses. Some methods are simple and transparent but cannot fully capture the nonlinear and coupled behavior of cascaded plants. Other methods are more flexible but difficult to tune and explain (Bernardes Jr et al., 2022; Mediavilla et al., 2022).

This paper focuses on using an AI-based approach that can learn the relationship between hydrologic–operational variables and plant generation directly from data and give a next-day forecast with improved accuracy (Avesani et al., 2022; Cohen, 2022; Elsheikh et al., 2021; Slater et al., 2023).

This is based on the concept of data-driven learning. Instead of writing down all physical equations and assumptions, the model observes many examples of past conditions—such as forebay elevation, inflow and outflow, spillway and governor gate positions, riparian flow, rainfall, and plant generation—and gradually learns a mapping from these inputs to the next-day generation (Amitaba et al., 2024; Balan et al., 2024; Bernardes Jr et al., 2022). The primary objective is to produce a forecasting model that represents the actual behavior of the Agus 1–7 cascade as closely as possible and is practical for operations use (Ji et al., 2018; Zhou et al., 2020).

In this paper, the idea of data-driven learning is implemented using an RBFNN. The “participating entities” in the model are the different input variables from the Agus plants and watershed. These variables play the part of signals or predictors that the RBFNN uses to build an internal representation of the system. After training, the network becomes a forecasting tool that can accept a new set of input conditions and produce a one-day-ahead generation forecast for each plant. The theoretical framework is therefore built around four main concepts: cascaded hydropower generation, forecasting, artificial neural networks, and the proposed RBFNN-based forecasting system (Schwenker et al., 2001; Sideratos & Hatziargyriou, 2020; Soper, 2023; Zhang et al., 2024).

1.6.1. Cascaded Hydropower Generation

Cascaded hydropower plants share the same river flow. Water released by an upstream plant becomes the inflow of the next downstream plant. For Agus 1–7, water comes from Lake Lanao, passes through each plant in sequence, and finally discharges to Iligan Bay. The generation of each plant depends on how much water passes through its turbines and the effective head between forebay and tailwater (Bernardes Jr et al., 2022).

Because of this arrangement, the plants cannot be treated as independent units. A

change in forebay elevation, gate opening, or spillage at one plant can change the inflow and head of the next plant. In addition, rainfall, riparian releases, and natural inflows affect lake level and river discharge. The theoretical view of this study is that the Agus cascade behaves as one dynamic, interconnected system, where generation at any time is the result of combined hydrologic conditions and operating decisions along the river (Ji et al., 2018; Moita & Monte, 2020; Rougé et al., 2019).

1.6.2. Forecasting

Forecasting is the process of making statements about future events whose actual outcomes have not yet been observed. In this study, the event of interest is the next-day hydropower generation of Agus 1–7. The forecast is expressed as a time series, such as hourly generation values for the next 24 hours (Mir et al., 2020; Prema et al., 2021).

Forecasting always involves uncertainty. The goal is not to predict the future perfectly but to provide an estimate that is as close as possible to what will actually happen, given the available data. For hydropower generation, this requires using up-to-date records of plant operation and hydrologic conditions. The more accurately the model can relate past and present variables to future generations, the more useful it becomes for day-ahead capacity nomination and scheduling (Akhtar et al., 2023; Essenfelder et al., 2020; Sapitang et al., 2020; Tsai et al., 2023).

1.6.3. Artificial Neural Networks and RBFNN

Artificial Neural Networks (ANNs) are computational models inspired by the way biological neurons process information. They consist of layers of interconnected nodes that transform input data into output predictions. When used for forecasting, an ANN is trained by repeatedly adjusting its internal parameters so that its outputs match the historical data as closely as possible (Baliyan et al., 2015; Bucci, 2020; SV et al., 2020).

An RBFNN is a particular type of ANN. It has three layers: an input layer, a hidden layer with radial basis functions (usually Gaussian), and an output layer. Each hidden neuron is “sensitive” to a certain region of the input space; it responds strongly when the

current inputs are close to its center and weakly when they are far away. By combining the responses of many such neurons, an RBFNN can approximate complex, nonlinear relationships (Sideratos & Hatziargyriou, 2020; Zhang et al., 2024).

In this thesis, the RBFNN is chosen because it is well suited for problems where the relationship between inputs and outputs is nonlinear but the forecasting horizon is relatively short. It can learn patterns in how generation responds to changes in forebay elevation, flows, gate positions, riparian releases, and rainfall without requiring an explicit physical model of all processes (Bernardes Jr et al., 2022; Schwenker et al., 2001; Soper, 2023).

1.6.4. The RBFNN-Based Forecasting System

Figure 1 (Architecture of the RBF neural network) illustrates the theoretical operation of the proposed system. Historical data from the Agus cascade—plant generation, forebay elevation, inflow and outflow, spillway and governor gate positions, riparian flow, and rainfall—are first collected and organized. These time-series data are then pre-processed to handle missing values, remove obvious errors, and scale the variables into suitable ranges for neural-network training (Cho et al., 2020; Kotsiantis et al., 2006; Maharana et al., 2022; Petrusevich, 2021).

After preprocessing, selected past values (lags) for these variables are fed into the RBFNN as input patterns. During training, the network adjusts the centers and weights of its radial basis functions so that the predicted generation for the next day matches the historical generation as closely as possible. Once training is complete, the model can accept a new set of recent measurements and produce a one-day-ahead generation forecast for each of the Agus plants.

The resulting system can then be evaluated using statistical accuracy measures such as MAPE (Chicco et al., 2021; De Myttenaere et al., 2016; Hodson, 2022). If the accuracy is acceptable, the forecasts may be used as support information for capacity nomination and operational planning.

1.6.5. Model Behaviour

The theoretical expectation is that the RBFNN will learn to mimic the typical behavior of the Agus cascade under different hydrologic and operating conditions. For example, when forebay elevation is high and inflows are strong, the model should predict higher potential generation, provided that gate positions indicate more water passing through the turbines. Conversely, when lake level is low or spillway flow is increased, the model should show a corresponding reduction in expected generation (Amitaba et al., 2024; Balan et al., 2024; Bernardes Jr et al., 2022; Fan et al., 2022).

Because the RBFNN uses many radial basis functions, it can represent different operating regimes: wet-season high inflow, dry-season low inflow, normal operation, or periods with unusual gate settings. In regions of the input space where many historical examples exist, the model is expected to give very accurate forecasts. In regions with fewer examples, the predictions may be less certain, which motivates continuous updating as new data become available (Sapitang et al., 2020; Zhou et al., 2020).

With this framework, the behavior of the model is intended to follow the actual behavior of the Agus cascade as closely as possible. When evaluated against historical data, the forecast curve of the trained RBFNN should closely track the observed generation curve. This match between forecast and actual behaviour is the main theoretical basis for using the RBFNN as a tool for short-term generation forecasting in the Agus 1–7 hydropower plants (Chicco et al., 2021; Kontopoulou et al., 2023).

Definition of Terms

Agus Cascade A series of interconnected hydroelectric power plants (Agus 1, 2, 4, 5, 6, and 7) where the outflow of upstream plants serves as inflow to downstream plants.

Radial Basis Function Neural Network (RBFNN) A type of artificial neural network that uses radial basis functions as activation functions, effective for modeling nonlinear relationships in time-series data.

Day-Ahead Forecasting The process of predicting system output, such as power genera-

tion, for the next 24-hour period based on current and historical data.

Operational MAPE A modified Mean Absolute Percentage Error that excludes low-generation values where percentage error becomes unstable or misleading.

Mean Absolute Error (MAE) A performance metric that measures the average magnitude of prediction errors without considering their direction.

Root Mean Square Error (RMSE) A metric that measures the square root of the average squared differences between predicted and actual values, emphasizing larger errors.

Coefficient of Determination (R^2) A statistical measure indicating how well the model explains the variability of the observed data.

Residual (Delta) Modeling A forecasting approach where the model predicts the change in output rather than the absolute value, improving continuity in time-series predictions.

Feature Engineering The process of creating input variables from raw data, including lag features, rolling statistics, and temporal indicators, to improve model performance.

Lag Features Past values of a variable used as inputs to capture temporal dependencies in time-series forecasting.

Rolling Statistics Statistical measures such as mean, standard deviation, minimum, and maximum computed over a moving window of time.

Planned Outage A scheduled shutdown of a generating unit, represented as a binary input (1 = available, 0 = unavailable) in the forecasting model.

Cascade Dependency The relationship between upstream and downstream plants in a hydropower system, where upstream discharge affects downstream generation.

Capacity Nomination The process of declaring the expected power generation output for a given period in power system operations.

Re-nomination The adjustment of previously declared generation schedules due to forecasting errors or operational changes.

Chronological Data Split A dataset division method where training, validation, and testing sets are separated based on time order to preserve temporal integrity.

Benchmark Models Alternative machine learning models (e.g., Random Forest and XGBoost) used for performance comparison against the proposed model.

Time-Series Data Sequential data points recorded over time, commonly used in forecasting applications.

Hydropower Generation The production of electricity using the energy of flowing or falling water.

CHAPTER 2

REVIEW OF RELATED LITERATURE

2.1. Short-Term Generation Forecasting in Cascaded Hydropower Systems

The liberalization of electricity markets worldwide has fundamentally repositioned storage hydropower producers, necessitating a shift in their operational objective from optimizing regional energy output to maximizing revenue through active market participation (Avesani et al., 2022). Hydropower remains a critical energy source, constituting approximately 15% of global electricity generation (Avesani et al., 2022). Critically, due to their inherent operational flexibility, storage hydropower systems are indispensable assets for maintaining energy balance and stability within national power grids, especially when facilitating the integration of intermittent Renewable Energy Sources (RES) like solar and wind power (Avesani et al., 2022).

The transition to a market-driven operational objective makes accurate short-term generation forecasting indispensable for two primary reasons: revenue maximization and grid stability. It enables producers to align water discharge schedules with high expected electricity price periods, optimizing revenue through a complex, non-linear process that is constrained by persistent hydrological uncertainties and the uncertain dynamics of the competitive electricity market (Avesani et al., 2022). At the same time, reliable forecasts are crucial for informed bidding in the day-ahead and intra-day markets, ensuring the timely commitment of flexible hydropower assets to support grid stability and maintain energy balance during periods of high demand or low intermittent RES output.

Cascaded hydropower systems significantly intensify the forecasting challenge. Unlike a single-plant system, a cascaded arrangement consists of multiple power plants situated sequentially along a river (Avesani et al., 2022). The power generation and water release decisions of an upstream plant directly influence the water inflow—and thus the potential generation—of all subsequent downstream plants. This creates a highly complex, coupled, non-linear system where the optimal operational schedule must account for the interdependencies and constraints of the entire chain, rather than just isolated units. Ac-

curate short-term forecasting must therefore model the dynamic hydraulic and economic relationships across the cascade.

Given the complexities and non-linear characteristics inherent in short-term generation forecasting, particularly in coupled hydro-econometric and cascaded environments, traditional statistical methods often lack the robust modeling capabilities required to maintain high prediction precision (Mediavilla et al., 2022).

As a result, there is substantial and growing research interest in the adoption of Artificial Intelligence (AI) and Machine Learning (ML) methodologies. AI models excel at identifying non-obvious correlations and complex, non-linear patterns within large, noisy datasets, frequently demonstrating superior accuracy over conventional models (Cohen, 2022; Elsheikh et al., 2021). Furthermore, the development of sophisticated hybrid forecasting models, which blend physics-based forecasts (such as those from numerical weather prediction) with data-driven AI/ML systems, is increasingly recognized as a vital approach for enhancing prediction skill and minimizing inherent biases (Slater et al., 2023).

The Radial Basis Function Neural Network (RBFNN) is a highly suitable Artificial Neural Network (ANN) architecture for this domain. It is distinguished by its capacity for effective generalization and simplicity in approximating nonlinear system behavior, even with a single hidden layer (Sideratos & Hatziaargyriou, 2020; Zhang et al., 2024). RBFNNs are efficiently utilized in hybrid schemes for tasks like input data clustering and localized regression, and their performance in short-term forecasting can be further optimized by combining them with meta-heuristic algorithms (Sideratos & Hatziaargyriou, 2020; Zhang et al., 2024).

2.2. Physical and Operational Characteristics of Cascaded Hydroelectric Power Plants

The foundation of hydropower optimization lies in understanding the conversion process: hydroelectric plants convert the kinetic and potential energy of flowing water into electrical energy (Bernardes Jr et al., 2022). Optimal power dispatch requires efficiently leveraging the available hydraulic head and river flows to maximize the resultant power,

which is primarily determined by the flow of water through the turbine and the effective height of the waterfall (Balan et al., 2024). Accurate short-term generation forecasting must therefore account for the gross head available, efficiency losses in the turbine–generator set, and hydraulic losses, often by integrating hydrological and climatic data to predict future water availability (Bernardes Jr et al., 2022).

The complexity of management is significantly increased in a cascaded hydro system, which features multiple power plants situated sequentially along the same river (Rougé et al., 2019). This setup creates a critical spatial coupling because the outflow (released water) from any upstream plant inherently becomes the inflow for its downstream counterpart (Moita & Monte, 2020). This interdependency means that upstream operational decisions—particularly release schedules—exhibit a progressive dominance over the entire chain, creating a highly non-linear system where localized downstream rules have limited influence (Rougé et al., 2019). Consequently, the core management challenge is the coordination of these interconnected facilities, which is governed by the mass balance principle (inflow minus outflow equals change in volume) and various operational thresholds.

Effective short-term forecasting and operation rely on monitoring and controlling several critical, coupled variables: reservoir volume and level, which must be maintained within predefined minimum and maximum storage limits and directly determine the available hydraulic head (Amitaba et al., 2024); inflow and outflow, where total outflow discharge consists of turbined flow that generates power and any intentional spillage (Balan et al., 2024); and turbine discharge and gate opening, which serve as key hourly decision variables that directly regulate power production and, in the context of flood management, dictate spillway operations (Amitaba et al., 2024).

These systems face practical coordination deficits, including managing uncertainty in future inflows, overcoming structural model deficits that fail to adequately represent inter-plant coordination, and mitigating the risk of strategic market behavior by upstream firms that might restrict water flow to limit downstream rivals (Moita & Monte, 2020; Rougé et al., 2019).

2.3. Hydropower Generation Forecasting and Time Horizons

The shift toward sustainable energy necessitates the accurate prediction of electricity generation across all resource types. Hydropower Generation Forecasting (HGF) involves estimating the electrical output of hydroelectric facilities over future time intervals, a process crucial for maintaining the operational reliability and economic viability of modern power systems (Akhtar et al., 2023). While HGF involves fewer direct meteorological uncertainties compared to highly intermittent sources like wind and solar, accurate short-term prediction remains critical for complex systems such as cascaded power plants to optimize water resource management and system integration. Forecasting techniques, whether applied to electricity load or generation, are conventionally stratified into distinct categories based on the temporal length of the predictive horizon, as these intervals correspond to specific planning and operational requirements (Mir et al., 2020). Long-Term Forecasting (LTLF) typically spans five years up to decades ahead, guiding strategic policy and power system expansion. Medium-Term Forecasting (MTLF) covers months to years ahead, supporting financial applications like energy trading and unit maintenance scheduling (Aquila et al., 2023; Prema et al., 2021). The most operationally relevant horizon is Short-Term Forecasting (STLF), characterized by timeframes ranging from a few hours up to one or two days ahead (Prema et al., 2021). This category is highly valuable for immediate system operation planning and coordination (Aquila et al., 2023).

The core focus of short-term generation forecasting in hydropower is essential for operational security, load-sharing decisions, and maximizing economic returns in the electricity market (Prema et al., 2021). Accurate STLF is vital for effective scheduling and dispatch, allowing operators to optimize system operations, minimize production costs, and avoid potential system overloads (Akhtar et al., 2023). Furthermore, STLF supports the integration of intermittent resources by providing the necessary predictive input for hydropower to maintain grid stability and compensate for fluctuations in wind and solar output. HGF is complicated because it involves many nonlinear physical and operational variables. These include hydrological inputs like rainfall and river inflow, reservoir metrics

like forebay elevation (water level), and operational controls like turbine discharge/outflow and gate openings (Balan et al., 2024; Bernardes Jr et al., 2022).

While hydropower is a conventional dispatchable source, the STLF challenge is significantly different from wind or solar, which struggle primarily with meteorological intermittency. For a cascaded hydroelectric system, the difficulties are rooted in hydraulic and spatial coupling. Specifically, the model must account for water routing time delays, which represent the highly variable and non-linear time it takes for released water to travel from an upstream dam to the reservoir of a downstream dam. It must also capture cascade effects, the precise interdependency where the outflow of one plant immediately and sequentially impacts the inflow and available head of the next plant in the chain (Ji et al., 2018).

2.4. Conventional Approaches and the Necessity of AI

Accurate forecasting of hydropower generation (HPG) is a fundamental prerequisite for the efficient management of water resources, capacity planning, and successful participation in dynamic energy markets (Hanoon et al., 2023). Historically, hydropower and related hydrological variables were forecasted using conventional statistical and physical models (Hanoon et al., 2023; Sapitang et al., 2020).

2.4.1. Traditional Forecasting Methods and Their Limitations

Among the classical techniques, the AutoRegressive Integrated Moving Average (ARIMA) model is a complex linear framework widely employed in time series analysis (Ghimire, 2017; Hanoon et al., 2023). Its seasonal counterpart, the Seasonal ARIMA (SARIMA) model, is applied when the time series exhibits clear periodic or seasonal patterns (Hanoon et al., 2023). Additionally, Multi-Linear Regression (MLR) and traditional hydrological/physical models represented the standard methodology used to forecast key variables like reservoir water levels (Sapitang et al., 2020).

Despite their mathematical simplicity and long-standing flexibility, these conventional methods face considerable limitations when applied to modern, complex hydroelec-

tric systems (Hanoon et al., 2023; Kontopoulou et al., 2023). A core difficulty is the restricted capability of linear models (such as MLR and ARIMA) to adequately model the significant non-linearity and non-stationarity found in hydrological and environmental time series (Kontopoulou et al., 2023). Complex, non-linear functions govern real-world systems, especially those related to coupled energy and water dynamics (Kontopoulou et al., 2023). Traditional techniques primarily focus on forecasting linear and static processes and struggle to identify these complex non-linear structures. Consequently, they exhibit poor performance, difficulty in accommodating multiple exogenous variables, and often fail to accurately generalize patterns, resulting in substantially higher prediction errors (Barzola-Monteses et al., 2019; Hanoon et al., 2023).

2.4.2. Application of AI in Energy Forecasting

The complexity arising from uncertain, dynamic, non-linear, and non-parametric variables in hydropower optimization necessitated a strategic shift toward AI and ML techniques (Bernardes Jr et al., 2022; Kontopoulou et al., 2023). ML models demonstrate preferable adaptability in prediction, circumventing the need for predefined classical mathematical definitions and efficiently modeling nonlinear processes (Sapitang et al., 2020). AI algorithms are now extensively applied across hydropower optimization, ranging from river flow forecasting to reservoir operation (Sapitang et al., 2020).

Key AI and ML models applied in energy and hydropower forecasting include:

Artificial Neural Networks (ANNs). The most widely utilized ML technique, deployed to predict river flow, reservoir inflow, and short- to medium-term production due to their robustness and superior capability to derive complex non-linear relationships (Bernardes Jr et al., 2022; Hanoon et al., 2023). The Radial Basis Function Neural Network (RBFNN) is a specific ANN architecture frequently investigated in forecasting literature and is the focus of this study (Hanoon et al., 2023).

Support Vector Machines (SVMs) and Regression (SVR). Widely applied due to their demonstrated efficacy in modeling non-linear processes and their superior performance over simpler linear methods for river discharge forecasting (Hanoon et al., 2023).

Tree-based Models. Algorithms like Random Forest (RF) and Boosted Decision Tree Regression (BDTR) constitute a significant category, successfully applied to forecasting and noted for their ability to handle smaller datasets and multiple exogenous inputs (Sapitang et al., 2020).

2.4.3. Performance Superiority of AI Models

AI models consistently demonstrate performance advantages over conventional statistical techniques in hydropower generation (HPG) time-series forecasting (Kontopoulou et al., 2023). This superiority stems from several key capabilities.

Modeling Non-linearity and Complex Dynamics. AI models, particularly Artificial Neural Networks (ANNs) and Support Vector Regression (SVR), exhibit superior capability in modeling the inherently non-linear processes of HPG data. This stands in contrast to traditional statistical techniques such as ARIMA, which often struggle when confronted with complex and highly dynamic structures (Hanoon et al., 2023; Kontopoulou et al., 2023).

Generalization and Accuracy. Machine learning approaches are consistently recognized as powerful tools that provide improved accuracy and generalization compared to conventional methods (Hanoon et al., 2023). Comparative studies frequently demonstrate that ANN and SVM models outperform ARIMA/ARIMAX techniques in predicting hydroelectric production across various forecasting horizons (Barzola-Monteses et al., 2022).

Adaptability to Complex Systems. The AI approach is generally preferred due to its stronger capability to model time-varying and complex systems. This is particularly important in cascaded hydroelectric power plants, where the operation of one reservoir non-linearly influences downstream units, creating highly coupled system dynamics (Bernardes Jr et al., 2022).

While deep learning models may sometimes require careful tuning of numerous hyperparameters, when properly optimized, AI algorithms generally achieve better prediction performance in most forecasting applications (Kontopoulou et al., 2023).

2.5. Artificial Neural Networks for Generation Forecasting

The increasing complexity and dynamic operation of modern power systems, particularly those incorporating renewable energy sources, have necessitated the development of highly accurate and adaptive forecasting methods (Baliyan et al., 2015). ANNs have emerged as one of the most effective families of models for handling the inherent non-linearity and complexity found in load and generation time series data (Baliyan et al., 2015; Bucci, 2020).

2.5.1. Conceptual Foundations and Suitability of ANNs

An ANN is defined as a highly connected array of elementary processors, or neurons, loosely inspired by the structure and distributed processing of the biological human brain (Baliyan et al., 2015). Fundamentally, an ANN functions as a non-parametric model designed to approximate the unknown function that generates observed data. The typical architecture includes an input layer, one or more hidden layers, and an output layer (Baliyan et al., 2015). The network is trained by iteratively updating the synaptic weights—which store the acquired knowledge—through algorithms like backpropagation, minimizing a cost function based on the output error (Baliyan et al., 2015). ANNs are exceptionally well-suited for short-term forecasting because they possess the capacity to approximate arbitrarily well a wide class of linear and nonlinear functions without requiring restrictive prior knowledge of the data-generating process (Bucci, 2020; Lu et al., 2022). This capability is critical when facing non-stationary time series and abrupt environmental or operational changes that render traditional statistical methods ineffective (Baliyan et al., 2015).

2.5.2. ANN Architectures and Applications in Energy Forecasting

The ANN family includes various architectures broadly divided into Feedforward Neural Networks (FFNNs), where signals propagate in a single direction (e.g., Multilayer Perceptron (MLP) and Radial Basis Function (RBF) networks), and Recurrent Neural Networks (RNNs), which incorporate internal feedback loops to retain state information and

model temporal dependencies (e.g., Long Short-Term Memory (LSTM) networks) (Lu et al., 2022). ANNs have been extensively employed across power system operations due to their ability to capture complex and non-linear relationships within time-series data.

Load and Renewable Forecasting. ANNs are among the most prominent non-parametric tools for short-term load forecasting (STLF) (Baliyan et al., 2015). Multilayer Perceptron Neural Networks (MLPNNs) have also been successfully applied to solar irradiance and rainfall forecasting by effectively modeling complex non-linear meteorological variables (Madhiarasan & Louzazni, 2022; SV et al., 2020).

Hydropower Prediction. ANNs have been successfully applied in numerous studies to predict hydroelectric power generation and reservoir inflow (Barzola-Monteses et al., 2022; Kaewarsa & Kongpaseuth, 2024). Furthermore, hybrid models that integrate ANNs with optimization algorithms have demonstrated superior accuracy in forecasting available energy, highlighting the adaptability and effectiveness of ANN-based approaches in hydropower systems (Kaewarsa & Kongpaseuth, 2024).

2.5.3. Advantages and Typical Limitations of ANNs

ANNs offer several key advantages over conventional statistical techniques in time-series forecasting applications.

Non-linear Modeling. The primary strength of ANNs lies in their ability to accurately approximate and model complex non-linear behaviors inherent in fluctuating and non-stationary time-series data, consistently producing more accurate results compared to traditional approaches (Barzola-Monteses et al., 2019; Bucci, 2020).

Data-Driven Generalization. ANNs are inherently data-driven and do not impose restrictive assumptions on the underlying data distribution, enabling them to implicitly detect and learn complex non-linear relationships between variables (SV et al., 2020).

Temporal Memory. Recurrent neural network architectures, such as RNNs and LSTM networks, utilize internal feedback mechanisms to capture and integrate historical dependencies into current predictions, a feature that is particularly crucial for time-series forecasting tasks (Lu et al., 2022).

Despite their advantages, ANNs also exhibit several limitations that must be considered in practical applications.

Data Requirements. ANNs are often described as “greedy” models because they typically require a substantial volume of historical data to train effectively and achieve reliable generalization (Bucci, 2020).

Overfitting and Complexity. Improper selection of network parameters can lead to overfitting, which significantly compromises the model’s ability to generalize to unseen data (Madhiarasan & Louzazni, 2022). Additionally, extensive trial-and-error is often required to determine the optimal network design, such as the appropriate number of hidden neurons (Madhiarasan & Louzazni, 2022).

Training and Computational Cost. Models with multiple hidden layers or complex recurrent structures, such as LSTMs, are inherently more complicated, require longer training times, and demand greater computational resources (Barzola-Monteses et al., 2019).

Hybrid modeling techniques and optimization algorithms often mitigate these limitations (Kaewarsa & Kongpaseuth, 2024). The suitability of the specific architecture, such as the Radial Basis Function Network, must be evaluated based on the complexity and volume of the available cascaded hydropower data.

2.6. The Radial Basis Function Neural Network (RBFNN)

The accurate short-term generation forecasting of cascaded hydroelectric power plants requires predictive models capable of handling complex, nonlinear system dynamics and rapidly processing large volumes of time-series data (Slater et al., 2023). Within the general ANN framework, the RBFNN architecture provides distinct, specialized advantages for high-precision function approximation tasks such as forecasting.

2.6.1. RBFNN Architecture and Differentiation from MLP

The RBFNN is a particular type of feed-forward network recognized as a universal function approximator, capable of approximating complex mathematical functions to any arbitrary degree of precision (Soper, 2023). This capability is shared with the widely used

Multi-Layer Perceptron (MLP) network. However, the RBFNN architecture fundamentally differs from conventional MLP networks in its structure and core function (Soper, 2023).

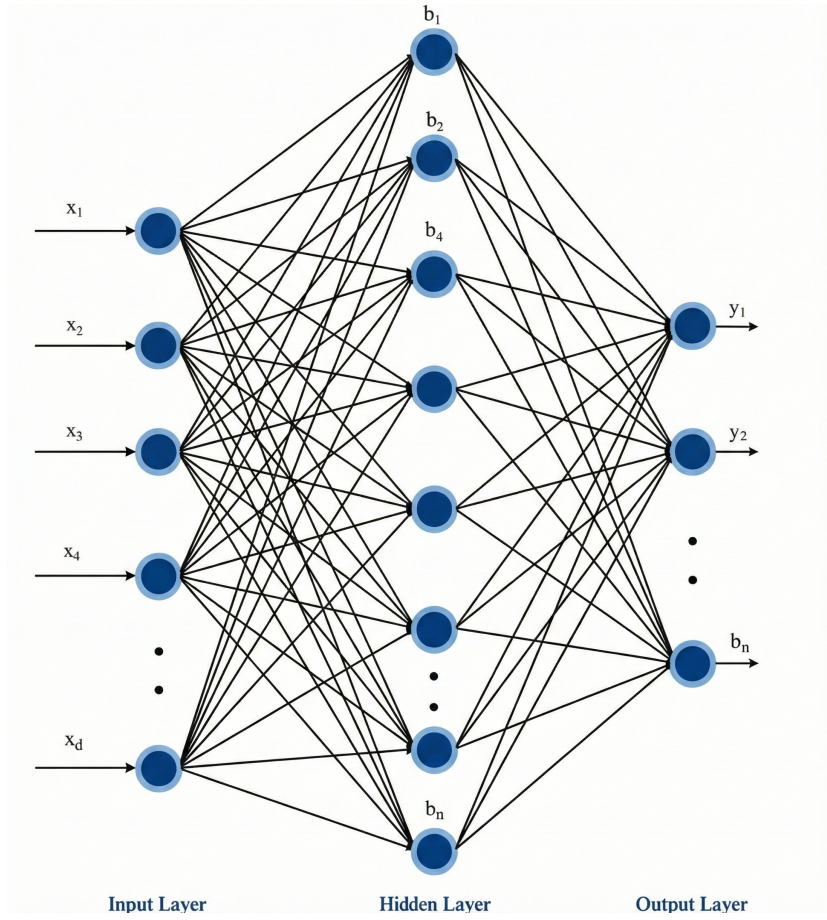


Figure 2.1. Architecture of the RBFNN(Chang et al., 2015).

The RBFNN relies on a simple three-layer structure: an input layer, a single hidden layer using non-linear radial basis functions (RBFs) as activation functions, and an output layer (Soper, 2023). In contrast, the MLP typically requires multiple hidden layers to achieve comparable high-precision function approximation (Soper, 2023). The core functional difference lies in the localized response of the hidden layer: each hidden neuron calculates a Gaussian function based on the Euclidean distance between the input vector and the neuron's stored "center" (Slater et al., 2023). This means RBFs function as local receptors, providing a strong output only when the input value is close to the node's center, differentiating it from the global response characteristic of MLP neurons (Soper, 2023).

2.6.2. Fundamentals of RBFNN Parameter Determination

The overall training task of an RBFNN is typically implemented through two-phase learning (Schwenker et al., 2001).

RBF Layer Parameters (Centers and Widths). The first phase involves determining the centers and widths or spreads of the RBF layer, often implemented through unsupervised learning (Montazer et al., 2018).

Centers: are vectors representing prototypes of the input data, and their choice is critical (Soper, 2023). Common methods for setting centers include random sampling or k-means clustering, which places centers in areas that best represent the overall data distribution (Soper, 2023). The *Widths (Spreads)* parameter controls the shape of the Gaussian basis function, determining the extent of the node's localized influence (Soper, 2023).

Output Weights. The second phase involves determining the weights between the RBF layer and the output layer (Soper, 2023). This is the only part of the RBFNN that requires supervised learning. Optimal weights can be determined either through iterative methods like gradient descent or, critically, by direct calculation using a pseudoinverse approach, which efficiently finds a minimum-norm weight solution for the given data (Soper, 2023).

2.6.3. RBFNN Success in Forecasting and Function Approximation

RBFNNs have proven versatile and powerful across a broad array of practical applications, confirming their utility in high-precision tasks (Soper, 2023). Successes span numerous domains, including the solution of partial differential equations, computer vision, and medical classification (Montazer et al., 2018).

For predicting dynamic systems, RBFNNs demonstrate strong performance in various time-series analysis and forecasting contexts:

Load Forecasting. RBFNNs have been successfully implemented for short-term load forecasting in electric power systems, demonstrating high accuracy in predicting loads 10 minutes ahead, an application that inherently deals with non-linearity and periodicity (Slater et al., 2023).

System Prediction. RBFNNs have been utilized for forecasting power in wind energy conversion systems and for categorizing high-resolution electrocardiograms (ECGs), demonstrating their effectiveness in processing 1D time-series data (Schwenker et al., 2001; Slater et al., 2023).

Complex Function Approximation. RBFNNs have undergone extensive testing on intricate non-linear mathematical optimization functions and real-world datasets, consistently producing precise predictions (Soper, 2023).

2.6.4. Suitability for Short-Term Hydro Generation Forecasting

The intrinsic characteristics of RBFNNs make them particularly well-suited for the specific demands of short-term generation forecasting in cascaded hydroelectric systems:

Fast and Efficient Training. The rapid training process of RBF networks—often orders of magnitude faster than MLPs—is a critical asset for short-term forecasting models that require frequent retraining or rapid iterative adjustment to reflect current operational conditions (Slater et al., 2023; Soper, 2023). Furthermore, RBF networks typically avoid the issue of being trapped in local minima often encountered in backpropagation-trained ANNs (Slater et al., 2023).

High Capacity for Non-Linear Dynamics. As universal function approximators, RBF networks excel at accurately approximating the highly complex, non-linear dynamics characteristic of cascaded hydro systems, which are governed by reservoir interactions and hydraulic constraints (Soper, 2023). This allows for crucial non-linear mapping between input stimuli and power output.

Local Approximation and Interpretability. The localized nature of the RBF functions allows individual nodes to specialize in approximating the function within a specific region of the input space (Soper, 2023). This local tuning helps the network effectively handle noisy or sparse data points common in real-world operational environments. Additionally, the RBF centers can often be interpreted as representative patterns of the data set, providing a degree of interpretability that is valuable in technical systems (Schwenker et al., 2001).

The combination of computational speed, powerful non-linear modeling capability, and localized learning makes the RBFNN architecture a robust and advantageous choice for the specific demands of short-term hydroelectric generation forecasting, prioritizing efficiency and accuracy in handling complex operational data.

2.6.5. RBFNN Training and Hyperparameter Tuning Methodology

The successful application of the RBFNN for short-term generation forecasting requires a rigorous training protocol focusing on effective parameter optimization and systematic hyperparameter tuning. The RBFNN's efficiency stems from its ability to employ hybrid learning where the hidden node parameters are set prior to optimizing the output weights (Aik et al., 2019).

A. Training RBFNN Parameters. The RBFNN training aims to find the optimal configuration of the centers, spreads, and output weights that minimize the Mean Square Error (MSE) (Horng et al., 2012).

Centers Selection and Clustering. The K-means (KM) algorithm is widely adopted for unsupervised center selection due to its speed and simplicity, grouping input cases into clusters where the centroids become the RBF centers (Aik et al., 2019). However, standard K-means is susceptible to the initial selection of centers and may not always extract the most accurate prototypes from complex time-series datasets, prompting the investigation of improved methods, such as the Distance-Weighted K-means (DWKM) algorithm (Aik et al., 2019).

Traditional and Gradient-Based Optimization. The Gradient Descent (GD) algorithm is a conventional derivative-based method often used for RBFNN training (Horng et al., 2012). However, like many derivative-based methods, GD frequently gets trapped in local minima and is generally outperformed by more advanced search techniques (Hamadneh, 2020).

Metaheuristic and Hybrid Optimization. To achieve global optimization and efficiently determine all complex parameters simultaneously, metaheuristic and swarm intelligence algorithms have become powerful alternatives (Hamadneh, 2020; Horng et al., 2012).

Algorithms like the Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and the Firefly Algorithm (FFA) model the entire set of RBFNN parameters (weights, spreads, centers) as a single solution vector. These techniques utilize randomized search to efficiently explore the solution space, demonstrating competitive or superior performance to traditional GD and GA algorithms when training RBFNNs for time-series prediction (Hamadneh, 2020)(Hamadneh, 2020).

B. Hyperparameter Tuning Strategies. Hyperparameter tuning focuses on selecting values for parameters that control the learning process, such as the number of hidden neurons (which equals the number of centers) and the spread parameter (σ)(Kahlout & Ekler, 2021).

Tuning via Experimentation and Validation. The most practical and common strategy involves extensive trial-and-error experimentation combined with the use of a validation dataset (Horng et al., 2012). Researchers typically test RBFNNs with varying numbers of hidden neurons to identify the configuration that yields the optimal performance metric on the validation set (Horng et al., 2012). This approach ensures the hyperparameter selection is robust and prevents overfitting to the training data.

Tuning via Optimization Algorithms and Criteria. In implementations utilizing metaheuristic training (PSO, FFA), the hyperparameter selection is often intrinsically part of the optimization process, as the algorithm directly searches for the best combination of centers and spreads alongside the weights (Horng et al., 2012). To formally evaluate the best resulting architecture, model selection criteria like the Schwarz Bayesian criterion (SBC) are utilized to assess the trade-off between minimal error and network complexity (Hamadneh, 2020). Cross-validation methods may also be used to ensure the chosen architecture is not biased towards a particular subset of the data (Horng et al., 2012).

2.6.6. Dataset Splitting in Time-Series Forecasting

In studies that use time-series forecasting, the dataset must be split up in strict chronological order so that the model only learns from the past to make predictions about the future. Unlike general machine learning problems, random splitting is inappropriate,

as it introduces future information into the training phase, leading to optimistically biased performance estimates.

The dataset is segmented into three distinct, non-overlapping subsets based on time: the training set (70%), which comprises the earliest data and is used for fitting the model parameters (centers, spreads, and weights); the validation set (15%), which follows the training set chronologically and is used to tune the model hyperparameters, such as selecting the optimal number of hidden neurons or spread value; and the testing set (15%), which consists of the most recent data and is reserved exclusively to provide a final, unbiased evaluation of the model's generalization capability on entirely unseen conditions (Kahlout & Ekler, 2021).

In quantitative studies, common ratios like 70% for training and 30% for testing (or 70–15–15% when a separate validation set is used) are frequently employed, provided this splitting is done sequentially (Hamadneh, 2020). This rigorous partitioning methodology is essential for developing a reliable short-term generation forecasting model for cascaded hydroelectric power plants.

2.7. Data Preprocessing and Quality Control

The performance, stability, and reliability of AI models, especially the RBFNN, are fundamentally contingent upon the volume, quality, and diversity of the input data (Maharana et al., 2022). Raw data collected from real-world applications are susceptible to numerous issues, necessitating robust preprocessing before being utilized for short-term generation forecasting (Maharana et al., 2022).

2.7.1. Data Problems in Hydropower Time-Series

Hydropower generation and associated meteorological and operational observations manifest as complex multivariate time-series data (Zhou et al., 2020). These datasets frequently exhibit intrinsic issues that complicate the forecasting process:

Missing Values and Irregular Sampling. Missing data are a common problem resulting from technical difficulties, sensor failures, or network disconnections (Petrusevich,

2021). These failures can create large gaps, sometimes stretching for weeks (Cho et al., 2020). Additionally, data acquisition may suffer from irregular sampling or asymmetric measurements due to transmission delays, complicating the time-alignment of observations (Zhou et al., 2020).

Sensor Noise and Inconsistency. Raw data can be affected by noise and corruption caused by how sensors work or how they communicate (Cho et al., 2020). Data inconsistency occurs when information gathered from various heterogeneous sources differs in standards or detail (Maharana et al., 2022). For sensitive machine learning models like ANNs, the presence of noisy and unreliable data significantly impedes the knowledge discovery process during training (Kotsiantis et al., 2006).

2.7.2. Strategies for Handling Missing Values and Imputation

Addressing missing data is critical because models that rely on continuous time series, such as RBFNNs, require input vectors that are complete and representative (Cho et al., 2020).

Simple Univariate Interpolation. For filling small data gaps, Linear Interpolation (LIN) is highly effective, assuming a linear function between the nearest good measurements (Cho et al., 2020). Spline Interpolation (SPL), such as using cubic splines, expands on this by creating a smooth, piecewise polynomial function that is less susceptible to high-variance periods (Petrusevich, 2021). Other basic methods include replacement using the mean, median, or mode (Maharana et al., 2022).

Advanced Model-Based and Multivariate Imputation. For handling larger gaps or complex patterns, more sophisticated methods are required. Multivariate approaches often yield superior results for large gaps by leveraging inherent periodicity and relationships within the dataset (Cho et al., 2020). These methods include K-Nearest Neighbors (KNN), which imputes values using a weighted mean of the k most similar data rows and has been shown to be accurate for medium gaps (Cho et al., 2020); Matrix Factorization (MF), which decomposes the time-series matrix to infer missing values and provides the most accurate imputations for large gaps across different sensor categories and missing rates (Cho et al.,

2020); and Autoregressive (AR) models, which estimate missing values as forecasts but require uninterrupted data segments for reliable coefficient estimation (Petrusevich, 2021).

2.7.3. Noise Reduction and Data Smoothing Techniques

Minimizing the presence of noise is essential prior to training sensitive AI models such as RBFNNs, as noise significantly impedes the knowledge discovery process (Maharana et al., 2022). Filtering and smoothing techniques improve trend detection and generalization performance by highlighting essential features (Hota et al., 2017).

Moving Average and Sliding Window. The Weighted Moving Average (WMA) is widely adopted for smoothing highly non-linear time series data, where the highest weight is assigned to the most recent data to enhance trend detection (Hota et al., 2017). This is often combined with a sliding window concept, where a fixed-size window aggregates historical data points to generate new, smoother input observations for the RBFNN (Hota et al., 2017).

General Filtering. Other methods include binning, which reduces the impact of minor observation errors by replacing binned values with derived general values (Maharana et al., 2022). More generally, sophisticated algorithms can be used that explicitly filter, modify, or relabel noisy data while preserving essential data features (Maharana et al., 2022).

2.7.4. The Crucial Role of Robust Data Preprocessing

Robust preprocessing is not merely a preliminary step but a critical determinant of the eventual success, stability, and reliability of AI-based forecasting models (Kotsiantis et al., 2006). Poor data quality significantly affects the accuracy and reliability of predictions, potentially leading to false conclusions (Maharana et al., 2022).

The substantial time dedicated to initial data preparation (estimated at 50% to 80% of the entire process) underscores its importance (Maharana et al., 2022). Data cleaning ensures the resultant dataset meets critical quality criteria, including validity, accuracy, completeness, consistency, and uniformity (Maharana et al., 2022). Ultimately, ensuring

that the training data is clean and representative maximizes the RBFNN's ability to extract meaningful non-linear information, thereby guaranteeing stable and reliable AI-based forecasts necessary for cascaded hydropower operation.

2.8. Evaluation Metrics

The validation of forecasting models requires robust statistical metrics to assess prediction performance. This study employs standard metrics commonly used across hydrological and energy forecasting literature: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination (R^2) (Chicco et al., 2021; De Myttenaere et al., 2016; Hodson, 2022).

Root Mean Squared Error is derived from the Mean Squared Error (MSE) and uses the L_2 -norm, involving the squaring of prediction errors (Chicco et al., 2021; Hodson, 2022). Because errors are squared, RMSE is inherently more sensitive to outliers, giving larger weights to points with high deviations (Chicco et al., 2021). Minimizing RMSE is statistically optimal if errors follow a normal (Gaussian) distribution (Hodson, 2022). RMSE yields a metric with the same units as the target variable (Hodson, 2022).

Mean Absolute Error minimizes the L_1 -norm, which uses a linear treatment of errors [Hodson, 2022]. MAE is recognized as a robust measure because the L_1 -norm tends to smooth out the penalty applied to extreme outliers, making it less sensitive to severe forecasting mistakes compared to RMSE (Chicco et al., 2021; Hodson, 2022). MAE is mathematically optimal when errors follow a Laplacian distribution (Hodson, 2022).

Mean Absolute Percentage Error is widely utilized and provides a highly intuitive interpretation in terms of relative error (Chicco et al., 2021; De Myttenaere et al., 2016). This metric is particularly useful when the application emphasizes sensitivity to proportional variations (De Myttenaere et al., 2016). A critical consideration is that MAPE is restricted to cases where the target variable is strictly positive and is known to be heavily biased towards low forecasts (Chicco et al., 2021).

Coefficient of Determination quantifies the proportion of variance in the dependent variable that is predictable from the independent variables (Chicco et al., 2021). R^2 offers a

bounded, scale-independent assessment of fit (Chicco et al., 2021), ranging from negative infinity up to 1 (perfect prediction). A value like $R^2 = 0.8$ clearly suggests a very good regression model performance, irrespective of the predicted generation output's scale (Chicco et al., 2021) .

These metrics are essential for a comprehensive evaluation, where MAPE is critical for assessing relative accuracy (desirable in generation forecasting) and R^2 provides an absolute measure of goodness-of-fit for comparing the RBFNN against alternative methods (Chicco et al., 2021; De Myttenaere et al., 2016).

2.9. Research Gaps

Despite the proven success of AI in general forecasting, several specific methodological and domain-focused gaps remain when approaching AI-based short-term generation forecasting for cascaded hydroelectric power plants. These gaps define the need for the present thesis:

Limited Focus on Cascaded Systems as a Forecasting Problem. The existing literature often treats cascaded hydropower as an optimization problem where forecasts are merely external inputs (Balan et al., 2024; Ji et al., 2018). Conversely, most hydropower forecasting studies focus on single-reservoir or individual station output, neglecting the core challenge of explicitly modeling the hydraulic coupling, routing delays, and head dependencies simultaneously across an entire cascade (Barzola-Monteses et al., 2019; Bernardes Jr et al., 2022). This leaves a gap in dedicated data-driven models for concurrent, coupled generation forecasting in cascaded systems.

Lack of Focused One-Day-Ahead Operational Resolution. While short-term horizons are recognized as critical, most hydropower forecasting is geared toward daily or longer resolutions for medium-term planning (Essenfelder et al., 2020; Sapitang et al., 2020). There is a clear lack of research explicitly targeting the one-day-ahead (24-hour) horizon with sufficient temporal granularity. This specific resolution is essential for crucial tasks like market bidding, spill management, and real-time coordination of reservoir releases, a gap highlighted by comparisons with advanced load and wind/solar forecasting

(Akhtar et al., 2023; Tsai et al., 2023).

Underutilization of RBFNNs. Despite their documented strengths as universal approximators with fast training speed, localized basis functions, and robustness in non-linear regression (Schwenker et al., 2001; Soper, 2023), RBFNNs are underutilized as the primary architecture for hydropower generation forecasting. Most studies favor MLPs, SVMs, or tree-based models (Hanoon et al., 2023; Kaewarsa & Kongpaseuth, 2024). This presents a gap in systematically applying and comparing the RBFNN architecture for one-day-ahead generation forecasting in complex cascaded systems.

Simple Feature Integration Lacking Cascade Physics. Many existing AI models for hydropower rely on relatively simple input sets. This often fails to fully exploit multivariate operational features that encode the physical coupling central to cascaded operation. Specifically, there is a gap in models that comprehensively integrate upstream releases, inter-reservoir travel time, and dynamic reservoir head across the cascade, which are critical physical drivers of short-term generation (Amitaba et al., 2024; Balan et al., 2024).

Preprocessing and Evaluation. A methodological gap exists concerning the documentation and consistent application of complete data quality pipelines. While the need for handling missing values, sensor noise, and chronological data splitting is stressed (Cho et al., 2020; Kahloot & Ekler, 2021), few studies provide a unified, reproducible preprocessing protocol tailored for multivariate cascaded hydropower data—combining advanced imputation, smoothing, sliding-window feature construction, and strictly chronological splitting for reliable time-series evaluation.

In response to these gaps, the present study develops an AI-based short-term (one-day-ahead) generation forecasting framework for a cascaded hydropower system using a Radial Basis Function Neural Network (RBFNN). The proposed model (i) treats the cascade as a coupled forecasting problem, (ii) explicitly incorporates cascade-relevant physical features such as upstream releases, travel time, and reservoir head, and (iii) employs a rigorous, time-consistent data preprocessing and evaluation protocol using MAPE. This framework is intended to serve as a practical and methodologically transparent benchmark

for operational forecasting in cascaded hydroelectric power plants.

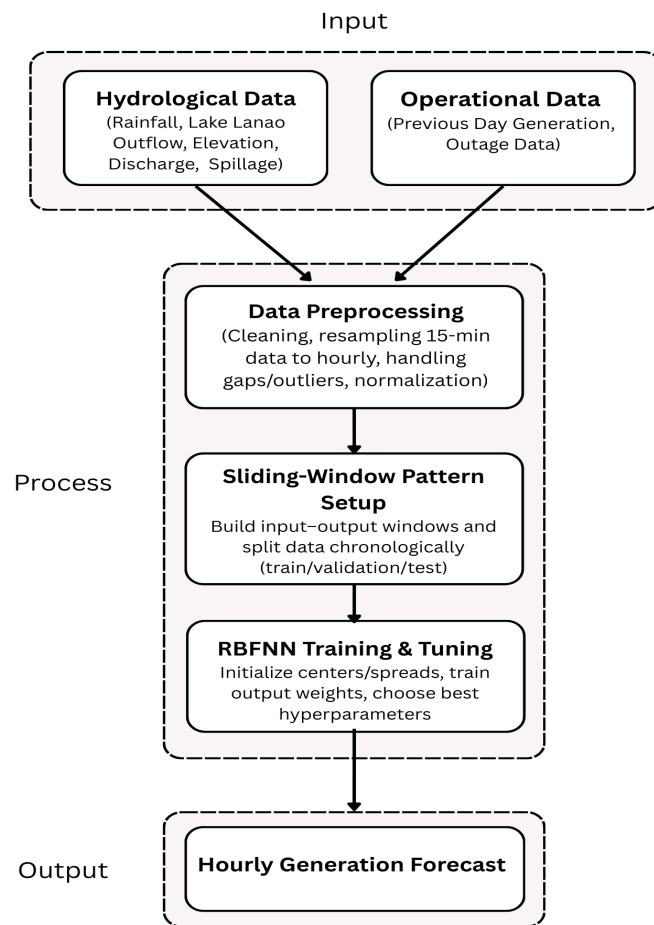


Figure 2.2. Conceptual Framework For Hourly Generation Forecast

CHAPTER 3

METHODOLOGY

This chapter presents the methodological framework of the study, where historical operational and hydrological data from the Agus 1–7 cascade were processed through a structured data-driven pipeline to develop a short-term forecasting system. The approach began with data collection, cleaning, and resampling into a consistent hourly dataset, followed by feature engineering that captured temporal patterns, cascade relationships, and unit availability through planned outages. These inputs were used to train a Radial Basis Function Neural Network (RBFNN) to learn the nonlinear behavior of the system and generate 24-hour ahead predictions for each plant and the total cascade. The model performance was then evaluated using statistical metrics and compared against benchmark models to assess its accuracy and operational suitability. The overall process is illustrated in Figure 3.1.

3.1. Research Design

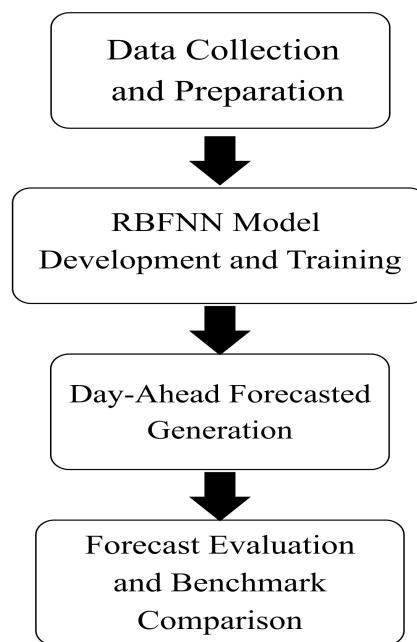


Figure 3.1. Workflow of the Study.

This study adopted a data-driven, supervised learning-based forecasting design, where historical operational data were used to train and evaluate a Radial Basis Function Neural Network (RBFNN) to model the nonlinear relationship between input variables and next-day hourly generation in the Agus 1–7 cascade.

RBFNNs were used because they have demonstrated strong performance in nonlinear time-series forecasting across power and water resources applications, including short-term PV power forecasting (Huang et al., 2015; Xu et al., 2019), short-term load forecasting (Chang et al., 2015), and rainfall prediction (El Shafie et al., 2012).

3.2. Data Collection and Processing

Data collection for this study was conducted using historical operational records obtained from NPC–MinGen, covering the Agus 1–7 hydropower cascade. The dataset included multiple operational variables describing the behavior of the cascade system, such as power generation, reservoir conditions, hydraulic flow, gate operations, environmental inputs, and unit availability. These data captured the nonlinear and time-dependent characteristics of hydropower generation. Since the dataset originated from a unified operational source, consistency and reliability across all variables were ensured without discrepancies arising from multiple data providers.

However, the collected data were recorded at varying temporal resolutions, including 5-minute, hourly, and daily intervals. To support the study objective of preparing a reliable dataset for short-term forecasting, all variables were standardized into a common hourly resolution. This unified structure served as the foundation for preprocessing, feature engineering, model development, and performance evaluation. A summary of the data sources used in this study is presented in Table 3.1.

Table 3.1. Summary of Data Sources

Data Type	Source	Resolution	Variables
Plant Generation	NPC–MinGen Operational Records	5-min (hourly aggregated)	Unit generation (MW), total generation per plant
Outage Data	NPC–MinGen Operational Records	Hourly (derived)	Unit availability (on/off status)
Reservoir Data	NPC–MinGen Operational Records	Hourly	Forebay elevation
Hydraulic Data	NPC–MinGen Operational Records	Daily (hourly upsampled)	Lake Lanao outflow
Gate Operations	NPC–MinGen Operational Records	Hourly	Spillway gate opening
Environmental Data	NPC–MinGen Operational Records	Daily (hourly upsampled)	Rainfall

3.2.1. Data Cleaning and Processing

The collected datasets underwent a structured preprocessing pipeline to ensure data quality, consistency, and suitability for machine learning modeling. Initial steps included cleaning and standardization of column names, conversion of variables into numerical formats, and reconstruction of datetime indices from separate date and time fields. Duplicate entries and invalid records were removed to maintain data integrity.

Consistency checks were performed using basic physical rules, such as non-negative generation values, acceptable reservoir levels, and continuity of flow-related variables. Records that violated these conditions were flagged and either corrected, if caused by obvious logging errors, or treated as missing values.

Missing values were handled using interpolation for short temporal gaps, while remaining missing entries were addressed using imputation methods. Outliers and abnormal spikes were identified and treated using smoothing and data-cleaning rules to preserve

physically meaningful behavior. Near-zero generation values were also carefully treated to distinguish between actual low production and outage conditions.

Unit outage status was not directly available in the dataset but was derived from generation data. Units with zero or near-zero generation were encoded as unavailable (0), while operating units were encoded as available (1). This binary representation allowed the model to incorporate operational constraints into the forecasting process.

3.2.2. Data Consistency and Temporal Alignment

To ensure coherence across all variables, all datasets were aligned to a common hourly timestamp index. Variables with different temporal resolutions were resampled accordingly: high-frequency plant generation data were aggregated from 5-minute intervals to hourly values, while lower-frequency variables such as Lake Lanao outflow and rainfall were upsampled from daily to hourly resolution using forward-filling.

This temporal alignment ensured that all input variables corresponded to the same time step, which is critical for multivariate time-series modeling. In cascaded hydropower systems, proper synchronization was essential because upstream and downstream interactions occurred over time. This standardization directly supported the study objective of transforming multi-resolution operational records into a consistent time-series dataset suitable for short-term forecasting.

3.2.3. Feature Engineering

To improve the predictive capability of the model, additional features were derived from the processed dataset. Temporal features such as hour-of-day, day-of-week, and seasonal indicators were included to represent periodic patterns in power generation. Lagged variables were generated to incorporate historical dependencies, enabling the model to learn from past system behavior.

Rolling statistical features, including moving averages and standard deviations, were computed to represent short-term trends and variability in the data. Operational features such as unit availability, derived from outage data, were also incorporated to

reflect real-world plant conditions.

Cascade relationships were explicitly modeled through upstream generation features, allowing downstream plants to utilize information from preceding plants in the cascade. This represents the physical dependency of water flow between interconnected plants and improves forecasting performance in cascaded hydropower systems.

These engineered features were selected based on their relevance to hydropower system dynamics and their ability to represent both temporal dependencies and physical interactions within the cascade, guided by domain knowledge and time-series analysis principles.

3.2.4. Normalization

To improve numerical stability and model convergence, all continuous variables were standardized prior to model training. Standardization (z-score normalization) was applied as follows:

$$x_{\text{std}} = \frac{x - \mu}{\sigma} \quad (3.1)$$

where x represents the original value, μ denotes the mean of the variable, and σ represents the standard deviation. The resulting standardized value x_{std} has a mean of zero and a standard deviation of one.

3.2.5. Input–Output Pattern Construction

The forecasting problem was formulated using a sliding-window approach, where input vectors were constructed from recent historical observations and the target corresponded to future generation values. Each input vector was defined as:

$$\mathbf{x}_i = [P_{i-1}, P_{i-2}, \dots, P_{i-L_P}, H_{i-1}, \dots, H_{i-L_H}, Q_{i-1}, \dots, Q_{i-L_Q}, \dots] \quad (3.2)$$

where P , H , and Q represent generation, reservoir level, and flow-related variables, respectively, and L denotes the selected lag lengths. The target output corresponded to future plant generation at a specified forecast horizon. The selection of lag lengths was guided by time-series analysis techniques and prior studies on short-term forecasting.

3.2.6. Dataset Splitting

To ensure realistic model evaluation and to prevent data leakage, the dataset was divided chronologically into three subsets: training, validation, and testing. The training set comprised 70% of the data and was used to fit the model parameters. The validation set, consisting of 15% of the data, was used for hyperparameter tuning and model selection. The remaining 15% was reserved as the testing set for final performance evaluation.

This chronological splitting approach ensured that the model was trained on past data and evaluated on future data, reflecting real-world forecasting conditions.

3.3. Research Instruments

The following software tools and development environments were used for data processing, model development, and system implementation. Microsoft Excel was utilized to collate, organize, and inspect the raw historical operational data, including generation, elevation, spillage, and gate opening records obtained from the plant operator. Python served as the primary programming language for data preprocessing, feature engineering, model training, and implementation of the complete forecasting pipeline. Key libraries used in the study included Pandas and NumPy for data manipulation, scikit-learn for preprocessing and evaluation, and TensorFlow/Keras for developing the Radial Basis Function Neural Network (RBFNN) model. Visual Studio Code was used as the integrated development environment (IDE) for code development, debugging, execution, and project management. All computations and model executions were performed in a local development environment to ensure reproducibility and efficient workflow management.

3.4. RBFNN Model Development

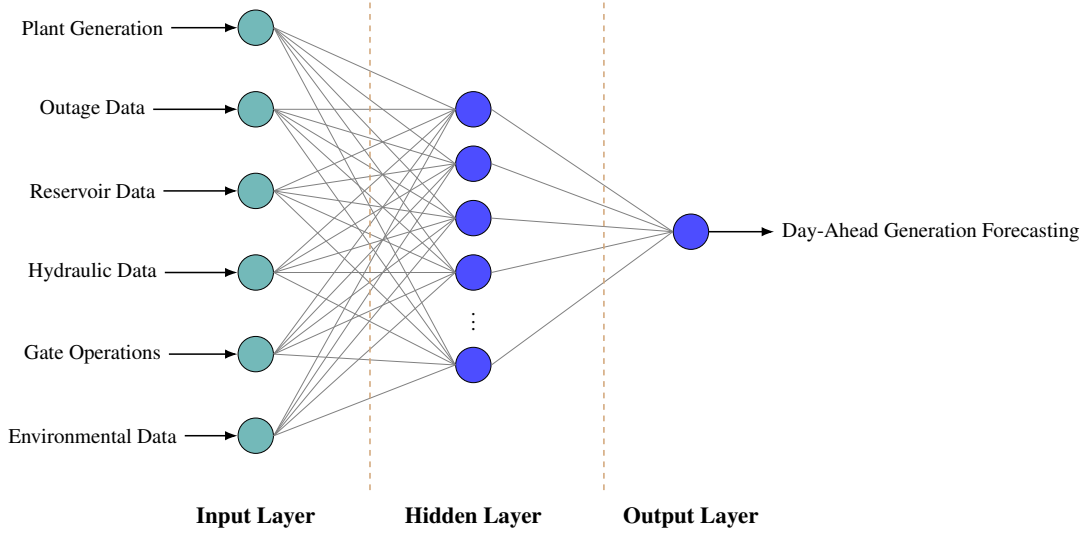


Figure 3.2. RBFNN architecture for day-ahead generation forecasting using operational and hydrological input variables(Chang et al., 2015).

The proposed forecasting model was based on a three-layer radial basis function neural network (RBFNN), consisting of an input layer, a hidden radial basis layer, and an output layer.

The input layer represented the feature vector constructed from the processed dataset. Each neuron corresponded to an input variable, including lagged generation values, reservoir levels, hydraulic variables, gate operations, rainfall, and derived operational features such as unit availability. The dimension m of the input vector was determined by the total number of selected features within the defined time window (Chang et al., 2015; Xu et al., 2019).

The hidden layer consisted of q radial basis neurons, where each neuron computed a Gaussian radial basis function defined as

$$\phi_j(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x} - \mu_j\|^2}{2\sigma_j^2}\right) \quad (3.3)$$

where μ_j represented the center and σ_j denoted the spread of the j -th basis function (Chang et al., 2015; Xu et al., 2019). These radial basis functions measure the similarity

between the input vector and predefined centers, allowing the network to capture nonlinear relationships in the data.

The output layer performed a linear combination of the hidden layer activations, expressed as

$$y_k(\mathbf{x}) = \sum_{j=1}^q w_{kj} \phi_j(\mathbf{x}) + \theta_k \quad (3.4)$$

where y_k initially represents the model output, which is interpreted as the predicted change in generation under the residual formulation, w_{kj} were the connection weights, and θ_k was the bias term. This structure enabled the model to approximate complex nonlinear mappings between system inputs and future generation values. Instead of directly predicting absolute generation values, the RBFNN was formulated as a residual (delta) model. In this formulation, the output y_k represents the predicted change in generation between consecutive time steps. The final forecasted generation was obtained by adding the predicted change to the most recent observed generation value. This approach improves model stability and reduces prediction drift, particularly under non-stationary hydrological conditions.

The overall architecture allowed the RBFNN to learn localized patterns through the radial basis functions while maintaining a simple linear mapping at the output layer. The number of hidden neurons and network configuration were selected empirically to balance model complexity and generalization performance, consistent with previous studies in RBFNN-based forecasting applications (El Shafie et al., 2012; Huang et al., 2015; Xu et al., 2019).

3.5. Model Training

The Radial Basis Function Neural Network (RBFNN) models were trained using a supervised learning approach on the preprocessed and feature-engineered hourly dataset derived from the Agus hydroelectric cascade system. Separate models were developed for each plant (Agus 1, Agus 2, Agus 4, Agus 5, Agus 6, and Agus 7) to capture plant-specific generation dynamics and cascade dependencies.

The input features included time-based variables, lagged generation values, rolling statistical features, hydraulic and environmental variables, as well as unit-level outage indicators and cascade-aware upstream features. Prior to training, all input variables and target outputs were standardized using z-score normalization to improve numerical stability and convergence.

The initialization of the radial basis layer was performed using K-Means clustering applied to the training dataset. This approach allowed the centers of the radial basis functions to be positioned effectively across the input feature space. The spread parameter of each radial basis function was treated as a trainable parameter and was adjusted during the learning process.

The model was trained to predict the change in generation between consecutive time steps rather than the absolute value. This residual (delta) formulation improved prediction stability by anchoring forecasts to the most recent observed generation values. This formulation was adopted to reduce prediction drift and improve model stability in non-stationary hydrological conditions.

Training was carried out using the Adam optimization algorithm with a learning rate of 0.0005, and the loss function was defined as Mean Squared Error (MSE). The training process was conducted for a maximum of 200 epochs with a batch size of 16. To prevent overfitting, early stopping was implemented based on validation loss with a patience of 15 epochs, restoring the best model weights when no further improvement was observed.

The dataset was divided chronologically into training (70%), validation (15%), and testing (15%) subsets to preserve the temporal structure of the data and prevent data leakage. The validation set was used to monitor model performance and guide hyperparameter selection, while the testing set was reserved for final evaluation.

During training, model performance was monitored using both training and validation loss curves to ensure convergence and detect potential overfitting. The trained models, along with their corresponding data scalers and configuration metadata, were saved for subsequent forecasting and evaluation stages.

3.6. Benchmark Models

To evaluate the effectiveness of the proposed Radial Basis Function Neural Network (RBFNN), two benchmark models were implemented for comparison, namely Random Forest (RF) and Extreme Gradient Boosting (XGBoost). These models were selected due to their widespread use and proven performance in regression and time-series forecasting applications.

Both benchmark models were trained using the same preprocessed dataset, feature set, and chronological data split as the RBFNN to ensure a fair and consistent comparison. The input variables included time-based features, lagged generation values, rolling statistical measures, operational variables, and outage indicators, consistent with those used in the main model.

The Random Forest model is an ensemble learning method based on multiple decision trees, where each tree is trained on a random subset of the data and features. The final prediction is obtained by averaging the outputs of all trees, which improves generalization performance and reduces overfitting.

The XGBoost model is a gradient boosting algorithm that builds trees sequentially, where each new tree attempts to correct the errors of the previous ones. This boosting process allows the model to capture complex nonlinear relationships and interactions among input variables.

Both models were trained using standard hyperparameter configurations and optimized using the validation dataset. The same evaluation procedure was applied across all models to ensure comparability of results. The performance of the RBFNN was then assessed relative to these benchmark models using multiple statistical metrics, as discussed in the subsequent section.

3.7. Evaluation Metrics

The performance of the proposed RBFNN and benchmark models was evaluated using multiple statistical metrics to assess prediction accuracy, error magnitude, and model

reliability. These metrics included Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination (R^2). The evaluation was conducted using the testing dataset to ensure an unbiased assessment of model performance.

The Mean Absolute Error (MAE) measures the average magnitude of prediction errors without considering their direction and is defined as

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3.5)$$

where y_i represents the actual value and $\hat{y}_i$ is the predicted value. MAE provides a direct measure of average error in the same units as the target variable (Hodson, 2022).

The Root Mean Squared Error (RMSE) evaluates the square root of the average squared differences between predicted and actual values and is given by

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3.6)$$

RMSE penalizes larger errors more heavily, making it sensitive to outliers and useful for assessing model robustness (Hodson, 2022).

The Mean Absolute Percentage Error (MAPE) expresses prediction error as a percentage and is defined as

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (3.7)$$

MAPE is widely used because it expresses accuracy in percentage form, making it easier to compare model performance across different scales and datasets (De Myttenaere et al., 2016). Operational MAPE was used to address the limitations of traditional MAPE in low-generation conditions, where percentage errors become unstable. A threshold based on plant capacity was applied to exclude near-zero actual values, ensuring meaningful evaluation of forecasting performance.

The Coefficient of Determination (R^2) measures the proportion of variance in the observed data explained by the model and is defined as

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3.8)$$

where $\bar{y}$ represents the mean of the observed values. A higher R^2 value indicates better model fit and explanatory power, and it has been shown to provide a more comprehensive evaluation of model performance compared to individual error metrics (Chicco et al., 2021).

The use of multiple evaluation metrics provided a comprehensive assessment of model performance by capturing different aspects of prediction accuracy. MAE and RMSE quantified absolute and squared errors, MAPE provided relative error interpretation, and R^2 evaluated the goodness of fit. This multi-metric evaluation ensured a robust comparison between the proposed RBFNN model and the benchmark models.

3.8. Forecasting Procedure

After the RBFNN models were trained and validated, they were used to generate day-ahead forecasts for the Agus 1–7 cascade. For a given forecast day ($D+1$), all required input variables, including recent hourly generation values, reservoir conditions, hydraulic variables, and operational indicators up to the end of day D , were prepared using the same feature engineering pipeline applied during training.

These input features were standardized using the scaling parameters derived from the training stage to ensure consistency in model input representation. The trained RBFNN models then produced hourly predictions for each plant over the 24-hour forecast horizon.

Instead of directly predicting generation levels, the model was formulated to predict generation changes (delta values) between consecutive time steps. The final generation output was obtained by adding the predicted change to the most recent observed value. This approach improved forecast stability and reduced drift under varying hydrological conditions.

Operational constraints were incorporated into the forecasting process through unit-level availability indicators derived from planned outage schedules. Plant-level predictions

were adjusted based on the proportion of available capacity relative to baseline conditions and subsequently distributed across active units. Units identified as unavailable (status = 0) were assigned zero generation, while available units followed the adjusted model predictions. Additional post-processing ensured that all outputs remained within physically feasible limits, including non-negativity and plant capacity constraints.

The final output of this stage consisted of hourly generation predictions for each plant, as well as the total cascade generation. These forecasts were then compared with actual measurements in the subsequent evaluation stage to assess model accuracy and operational applicability.

This ensured that the forecasts were both data-driven and consistent with real-world operational constraints of the hydropower system.

3.9. Data Management Plan (DMP)

The study implemented a Data Management Plan (DMP) to ensure that all datasets, scripts, and results were well organized, reproducible, and protected throughout the thesis lifecycle. The DMP was not treated as mere documentation; instead, it served as a practical framework guiding how the Agus cascade data were handled from initial acquisition to long-term archiving.

Data organization was achieved through the use of descriptive file naming conventions and a structured folder hierarchy, allowing every figure, table, and result to be traced back to its corresponding dataset and processing script. A README file was also maintained to document the purpose and contents of each directory, ensuring clarity and ease of navigation.

To ensure reproducibility, all preprocessing, model training, and evaluation procedures were implemented in well-documented scripts with clear annotations and fixed random seeds where applicable. The DMP recorded software versions, key hyperparameters, and the exact train-validation-test splits used in the experiments. This provided a complete roadmap that enabled the adviser, examiners, and future researchers to replicate the RBFNN forecasting workflow and verify or extend the results.

Data protection measures were also established to safeguard the integrity of the study. All working data and code were stored on a password-protected system, with regular backups maintained on institutional drives or encrypted cloud storage. Backup schedules and recovery procedures were defined to mitigate risks associated with hardware failure or accidental data loss. Access to the full dataset was restricted to the researcher and adviser, and any shared data were anonymized or aggregated in compliance with institutional and utility confidentiality requirements.

This ensured reproducibility of the RBFNN forecasting pipeline and consistency of the reported results.

Overall, this methodological framework ensured that the forecasting system was developed, trained, and evaluated in a consistent and reproducible manner, enabling reliable short-term generation prediction for the Agus cascade system.

CHAPTER 4

RESULTS AND DISCUSSION

4.1. Overview of Results

This chapter presents the results of the proposed Radial Basis Function Neural Network (RBFNN) forecasting model applied to the Agus 1–7 hydropower cascade. The performance of the model is evaluated using multiple statistical metrics and compared with benchmark models, including Random Forest and XGBoost.

The results are organized to reflect the overall modeling pipeline, beginning with model training behavior, followed by forecasting performance, benchmark comparison, and detailed error analysis. In addition, the operational behavior of the model under cascade interactions and planned outages is examined to assess its practical applicability.

The findings of this study are discussed in relation to the objectives outlined in Chapter 1, highlighting the effectiveness of the proposed approach in improving short-term hydropower generation forecasting.

4.2. Data Preparation Results

4.2.1. Dataset Overview

The cleaned and processed dataset was derived from historical operational records of the Agus hydropower cascade. All variables were standardized into a uniform hourly time resolution to ensure consistency across inputs.

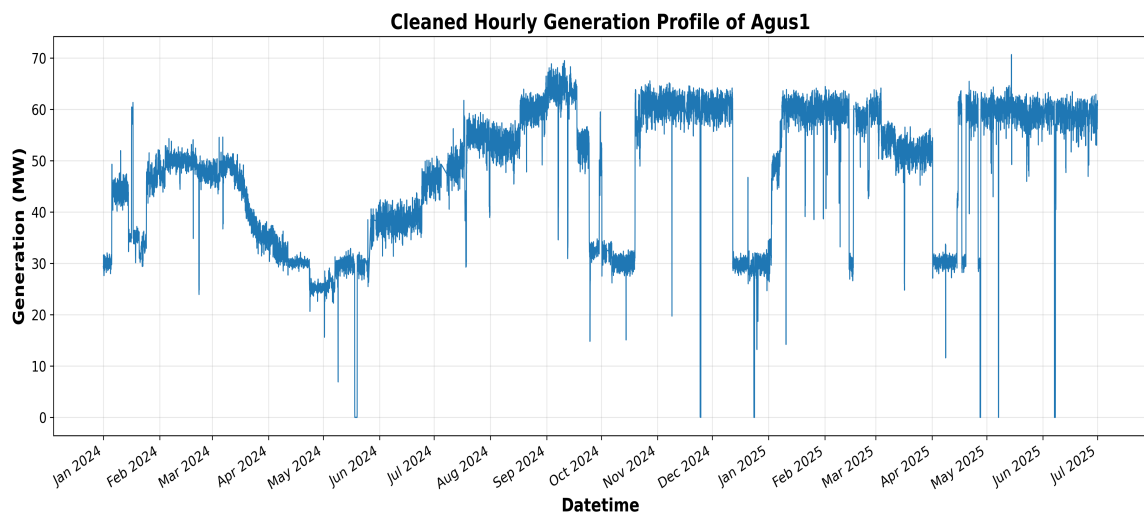
Table 4.1. Summary of Processed Dataset

Description	Value
Time Range	01/01/2024 – 06/30/2025
Total Rows	13131
Number of Features	53
Resolution	Hourly

Table 4.2. Input Variables Used in the Forecasting Model

Input Variable Group	Description
Plant Generation	Hourly unit and total generation per plant.
Reservoir Variables	Forebay elevation.
Hydraulic Variables	Lake Lanao outflow.
Gate Operation Variables	Total spillway gate openings.
Outage Indicators	Unit availability (1 = on, 0 = off).

4.2.2. Cleaned Data Visualization

**Figure 4.1.** Cleaned hourly generation profile of Agus1.

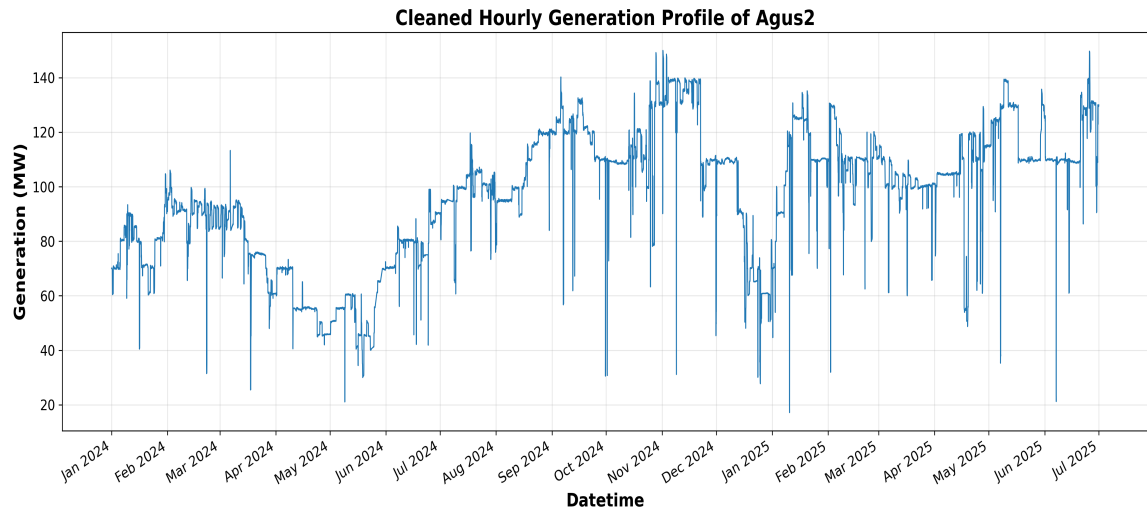


Figure 4.2. Cleaned hourly generation profile of Agus2.

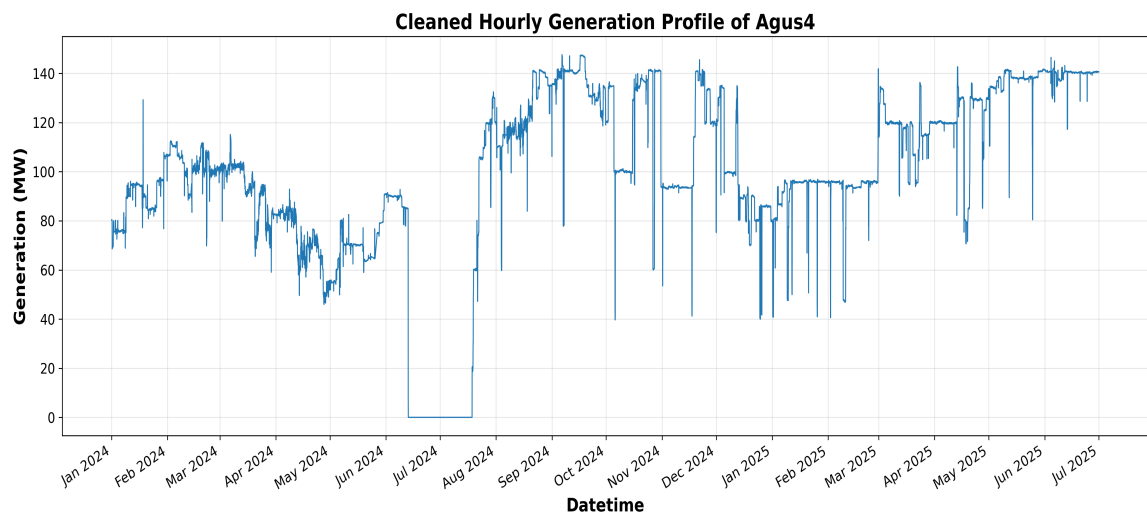


Figure 4.3. Cleaned hourly generation profile of Agus4.

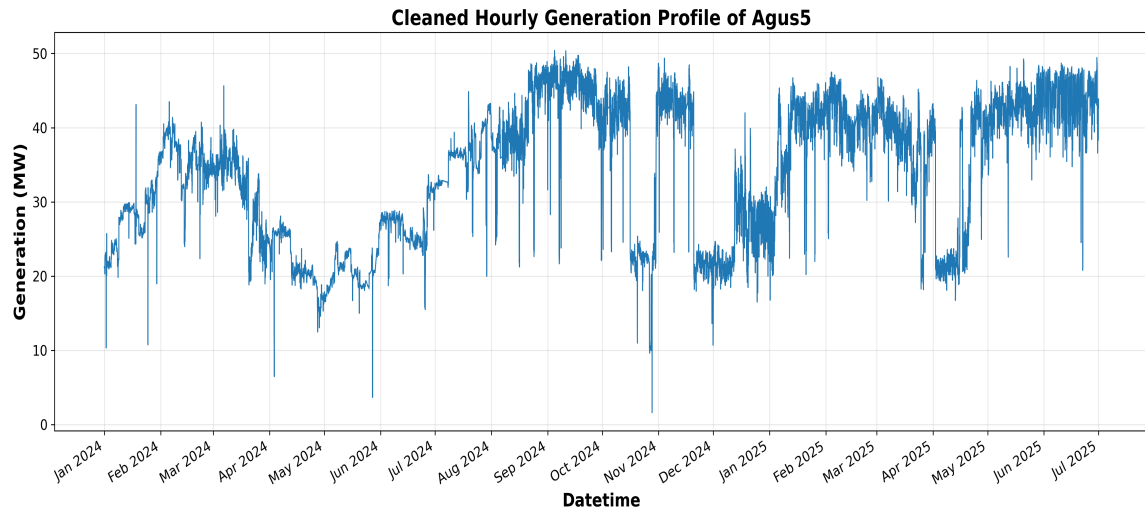


Figure 4.4. Cleaned hourly generation profile of Agus5.

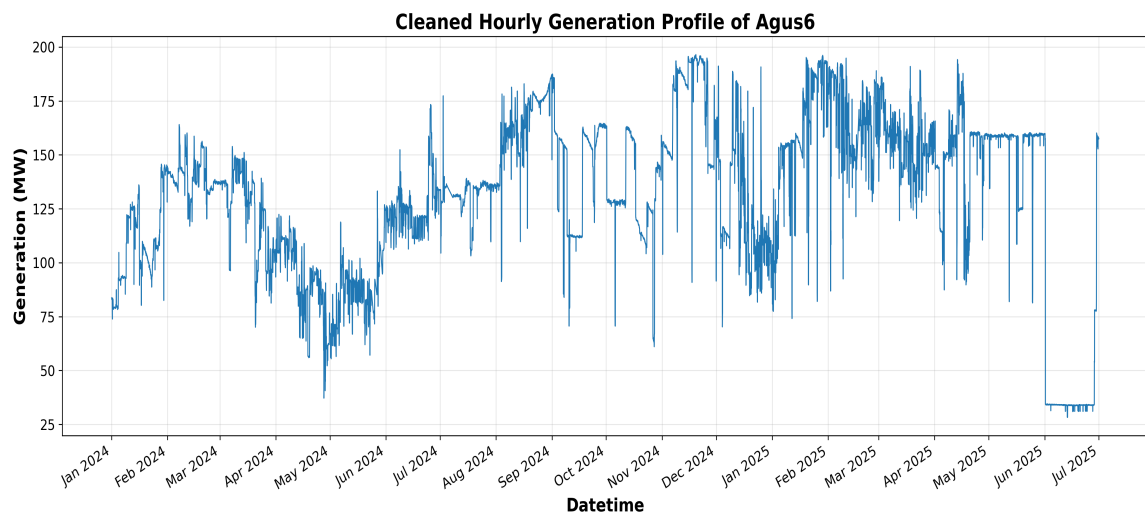


Figure 4.5. Cleaned hourly generation profile of Agus6.

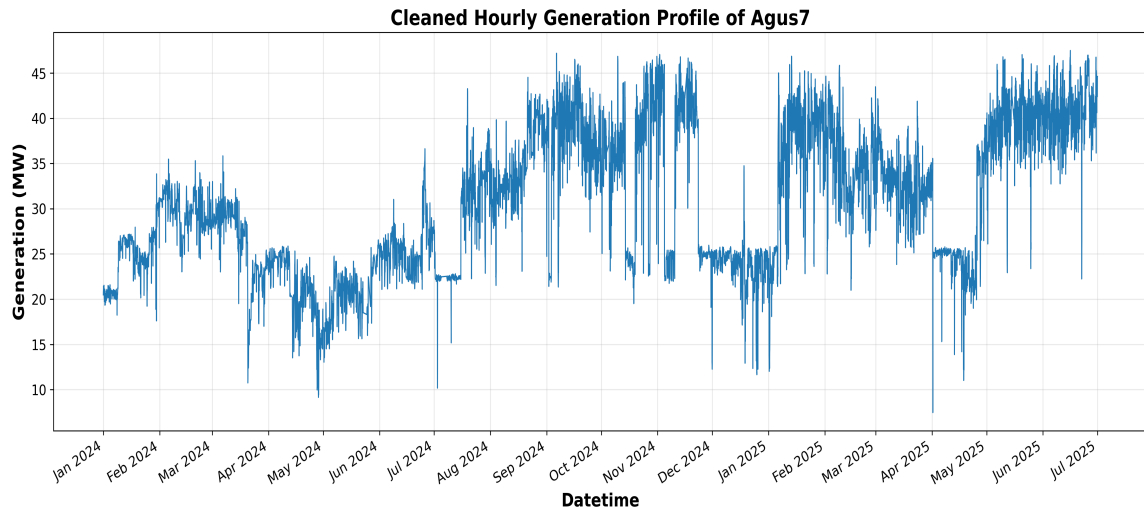


Figure 4.6. Cleaned hourly generation profile of Agus7.

4.2.3. Discussion of Cleaned Data

The cleaned hourly generation profiles presented above illustrates the operational behavior of each plant in the Agus hydropower cascade after preprocessing. The results confirm that the applied data cleaning procedures successfully produced continuous and consistent time series suitable for modeling.

Across all plants, the generation profiles exhibit expected temporal variability driven by operational scheduling, hydraulic conditions, and demand patterns. Upstream plants such as Agus 1 and Agus 2 show relatively smoother generation trends, while downstream plants, particularly Agus 5, Agus 6, and Agus 7, display higher variability. This behavior is consistent with the cascading nature of the system, where downstream generation is influenced by upstream discharge and operational adjustments.

Notable fluctuations and sudden drops in generation are observed in several plants. These are attributed to operational events such as unit outages, maintenance schedules, or abrupt changes in water release. The presence of such variations indicates that the dataset captures realistic plant behavior rather than being overly smoothed, which is essential for training robust forecasting models.

Additionally, the absence of prolonged missing segments and the continuity of the time series confirm that the interpolation and data repair techniques were effective. Minor

spikes and short-term irregularities remain in the data, reflecting real-world measurement noise and operational dynamics, which are important for improving model generalization.

Overall, the cleaned dataset preserves the inherent temporal patterns and interdependencies of the Agus cascade while ensuring data quality and consistency. This provides a reliable foundation for subsequent feature engineering and model training stages.

4.3. RBFNN Training and Validation Results

4.3.1. Model Training and Validation Results

This section presents the training and validation performance of the proposed Radial Basis Function Neural Network (RBFNN) models applied to the Agus hydropower cascade. Separate models were developed for each plant, namely Agus 1, Agus 2, Agus 4, Agus 5, Agus 6, and Agus 7, to capture plant-specific operational characteristics.

The evaluation focuses on three key aspects: (1) the selected model configuration and calibration parameters, (2) validation performance based on standard error metrics, and (3) training behavior assessed through loss curves. These results provide insight into the capability of the RBFNN models to learn the nonlinear relationships present in the hydropower system and generalize to unseen data.

The optimized hyperparameters and calibration settings used for each plant are summarized in Table 4.3. These parameters were selected to balance model accuracy and stability across the different plants in the cascade.

Table 4.3. RBFNN Model Configuration and Calibration Parameters per Agus Plant

Plant	Feature Count	Centers	Learning Rate	Shrinkage	Bias Correction (MW)	MAPE Threshold (MW)
Agus 1	158	80	0.0010	1.15	-0.0058	1.000
Agus 2	169	120	0.0005	1.15	0.0027	1.800
Agus 4	169	180	0.0005	1.15	0.0019	1.581
Agus 5	168	180	0.0010	1.15	-0.0131	1.000
Agus 6	171	80	0.0005	1.15	-0.0089	2.190
Agus 7	168	120	0.0010	1.15	0.0075	1.000

Table 4.4 presents the validation performance of the RBFNN models for each Agus plant. The models were evaluated using Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2).

Table 4.4. RBFNN Validation Performance per Agus Plant

Plant	MAPE (%)	MAE (MW)	RMSE (MW)	R^2
Agus 1	3.471	1.727	2.756	0.929
Agus 2	1.080	0.984	4.297	0.822
Agus 4	0.827	0.717	3.362	0.953
Agus 5	4.091	1.469	2.231	0.904
Agus 6	2.413	3.667	7.981	0.835
Agus 7	3.806	1.234	2.109	0.824

To provide an overall assessment of model performance, the average validation metrics across all plants are summarized in Table 4.5. These aggregated results represent the general forecasting capability of the RBFNN model for the Agus cascade.

Table 4.5. Overall Average Validation Performance of the RBFNN Model

Plant	MAPE (%)	MAE (MW)	RMSE (MW)	R^2
Agus Cascade	2.615	1.633	3.789	0.878

The training performance of the RBFNN models was evaluated using training and validation loss curves, as shown in Figures 4.7 to 4.12. These curves were used to assess convergence behavior and generalization capability during model training.

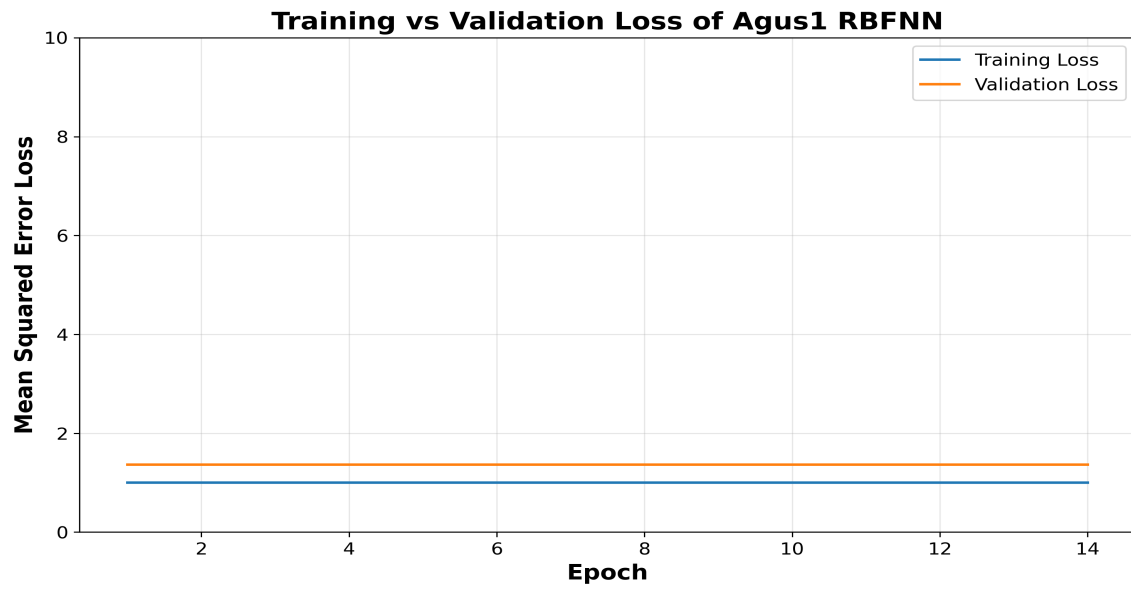


Figure 4.7. Training and validation loss curves of the RBFNN models for Agus 1.

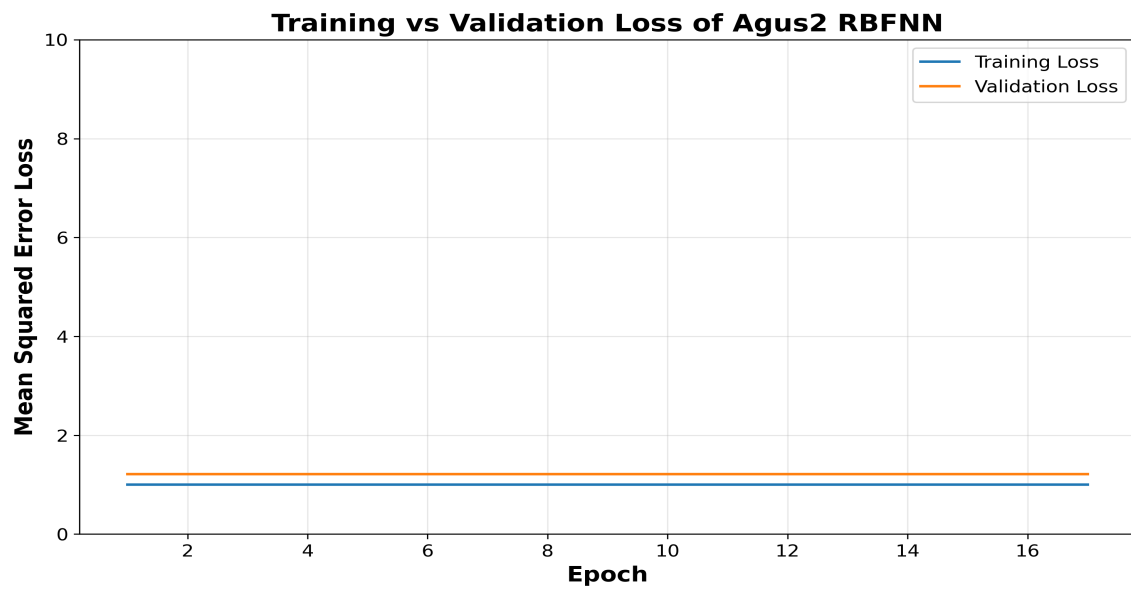


Figure 4.8. Training and validation loss curves of the RBFNN models for Agus 2.

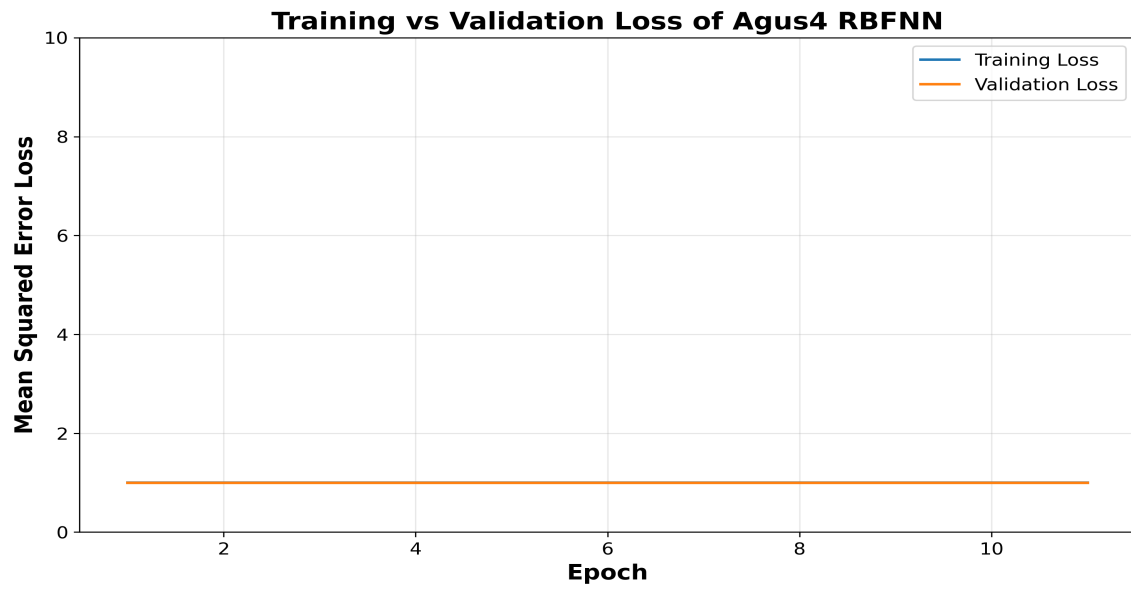


Figure 4.9. Training and validation loss curves of the RBFNN models for Agus 4.

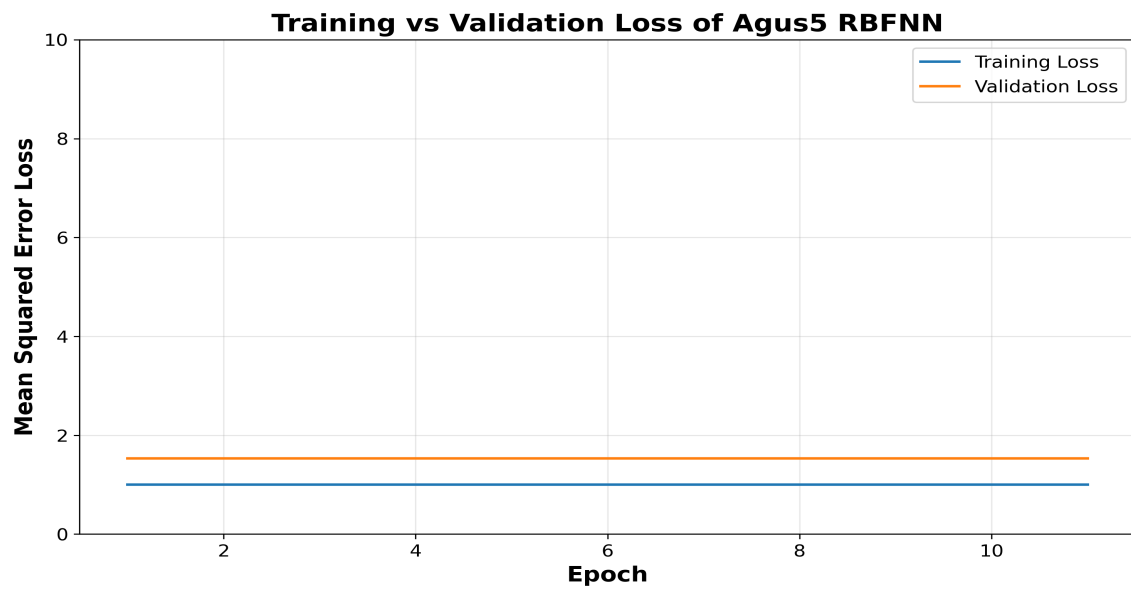


Figure 4.10. Training and validation loss curves of the RBFNN models for Agus 5.

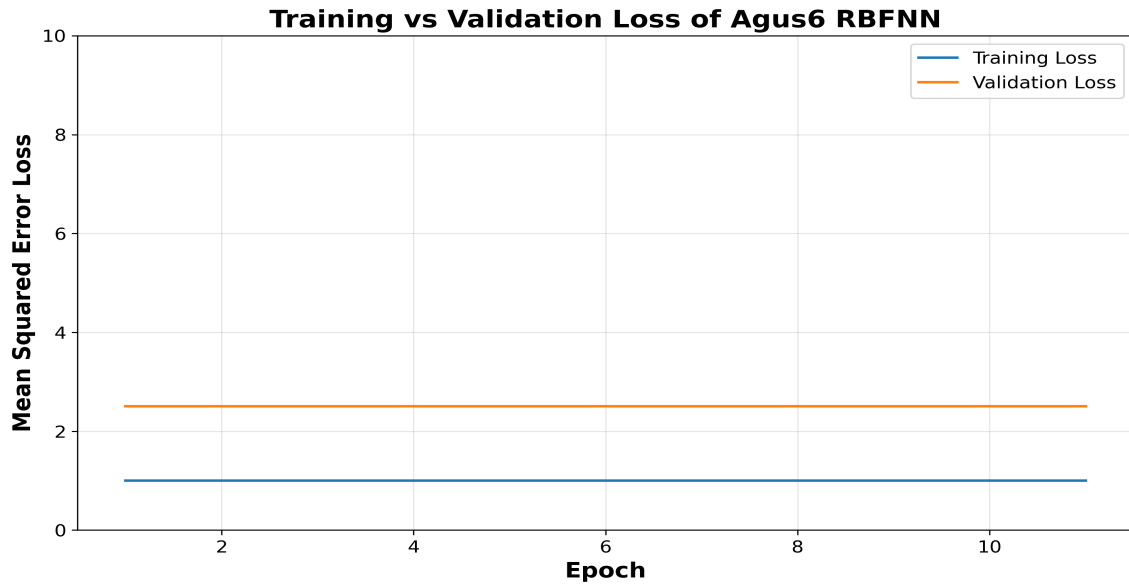


Figure 4.11. Training and validation loss curves of the RBFNN models for Agus 6.

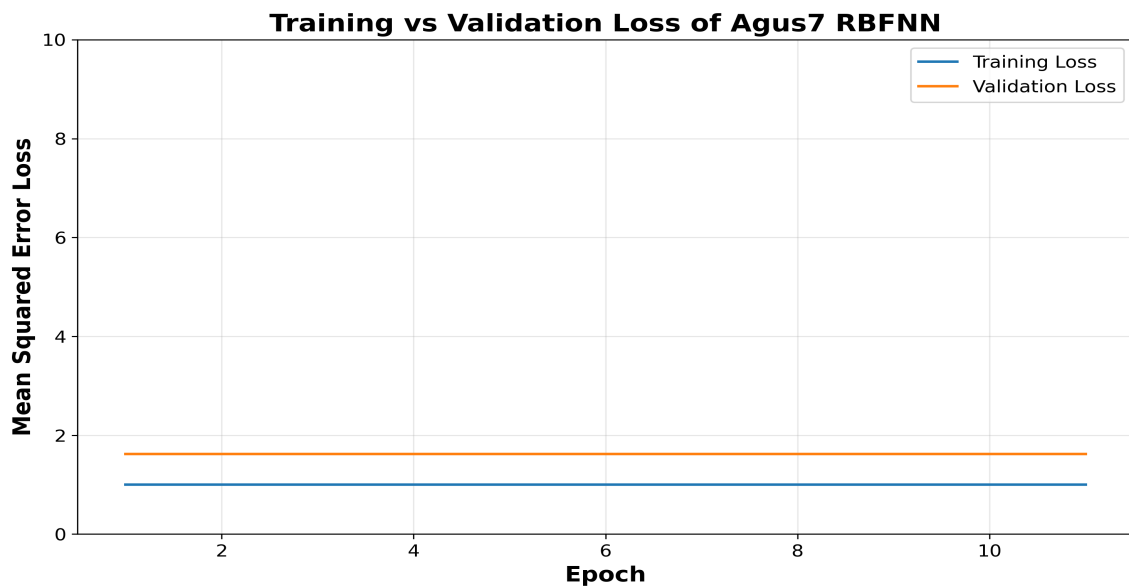


Figure 4.12. Training and validation loss curves of the RBFNN models for Agus 7.

The training and validation loss curves appear relatively flat across epochs, indicating rapid convergence rather than the absence of learning. This behavior is expected due to the structure of the RBFNN and the nature of the prediction target.

Specifically, the model was trained to predict short-term changes in generation rather than absolute values. Since hydropower generation typically varies smoothly over

time, the target values exhibit low variance and are centered near zero after normalization. As a result, the loss function stabilizes quickly and shows minimal variation during training.

Furthermore, the loss values were computed on standardized data, which reduces the magnitude of error and contributes to the observed stability of the curves. Therefore, the loss curves alone do not fully reflect model performance in physical units.

The effectiveness of the model is better demonstrated through validation metrics such as MAPE, MAE, RMSE, and R^2 , which indicate strong predictive accuracy across all Agus plants.

4.4. Forecasting Performance of the RBFNN Model

The forecasting performance of the RBFNN model was evaluated using the testing dataset. The model generated day-ahead hourly predictions for each plant in the Agus cascade, as well as the total cascade generation.

The results indicate that the RBFNN model was able to closely follow the actual generation patterns, effectively capturing both short-term fluctuations and overall trends. The use of a residual (delta) formulation improved forecast stability by anchoring predictions to recent observed values.

Tables 4.6 and 4.7 summarize the testing performance of the model in terms of MAPE, MAE, RMSE, and R^2 .

Table 4.6. Overall Testing Performance of the RBFNN Model

Plant	MAPE (%)	MAE (MW)	RMSE (MW)	R^2
Agus Cascade	2.519	1.355	3.575	0.925

Table 4.7. RBFNN Testing Performance per Agus Plant

Plant	MAPE (%)	MAE (MW)	RMSE (MW)	R^2
Agus 1	3.476	1.841	3.440	0.899
Agus 2	1.372	1.164	4.810	0.912
Agus 4	0.611	0.697	3.077	0.945
Agus 5	4.040	1.544	2.189	0.901
Agus 6	1.405	1.396	5.679	0.990
Agus 7	4.209	1.491	2.254	0.905

The results show that the model maintained strong predictive accuracy on unseen data, achieving an average testing MAPE of 2.519% across the Agus cascade. This indicates low percentage forecasting error and reliable generalization performance.

At the plant level, Agus 4 achieved the lowest testing MAPE, indicating the highest prediction accuracy, while Agus 7 exhibited comparatively higher variability. This higher error in Agus 7 may be attributed to increased operational variability and sensitivity to upstream cascade conditions, which makes accurate forecasting more challenging compared to other plants. Despite this, all plants achieved high R^2 values, with Agus 6 reaching the highest value of 0.990, demonstrating that the model effectively captured the variability in hydropower generation.

Overall, the combination of low error metrics and high coefficient of determination confirms that the RBFNN model generalized well to unseen data and is suitable for short-term hydropower generation forecasting in the Agus cascade.

4.4.1. Actual vs Forecast Results

The quantitative evaluation presented in the previous subsection is further supported through visual analysis of actual and forecasted generation profiles.

This subsection presents the comparison between the actual and forecasted generation values of the RBFNN model for each Agus hydropower plant. The analysis is based

on the testing dataset, which consists of unseen chronological data not used during model training and validation.

The purpose of this comparison is to visually evaluate the ability of the model to capture the temporal behavior of hydropower generation, including both short-term variations and overall operational trends.

Figures 4.13 to 4.18 illustrate the comparison between actual and forecasted generation of the RBFNN model for all Agus plants during the testing period.

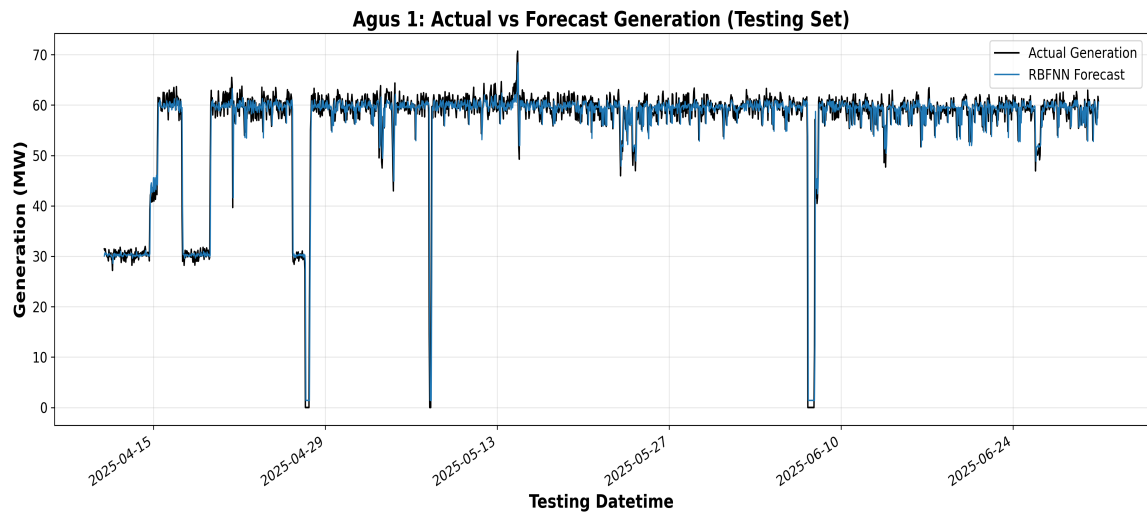


Figure 4.13. Actual and forecasted generation of Agus 1 during the testing period.

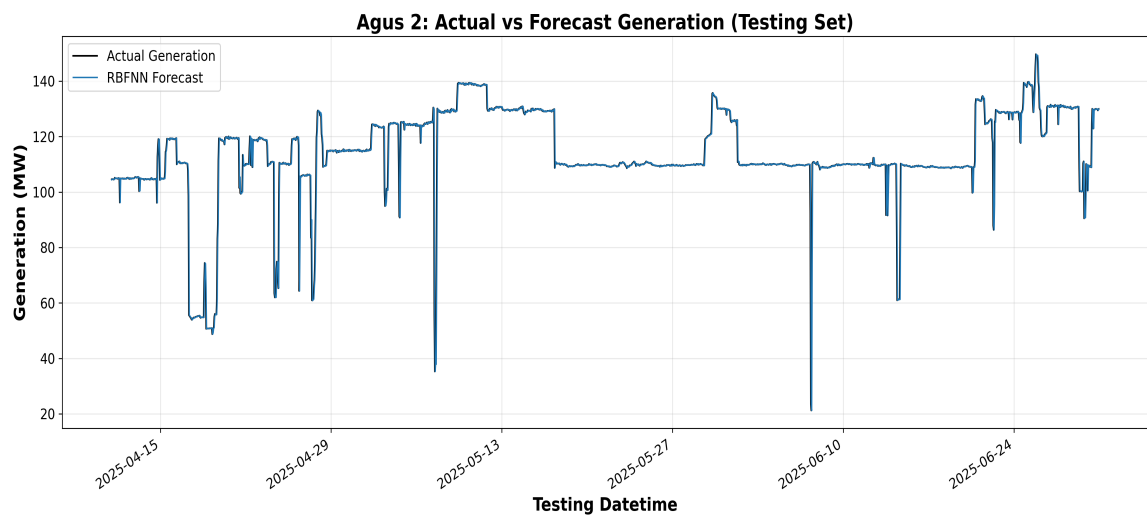


Figure 4.14. Actual and forecasted generation of Agus 2 during the testing period.

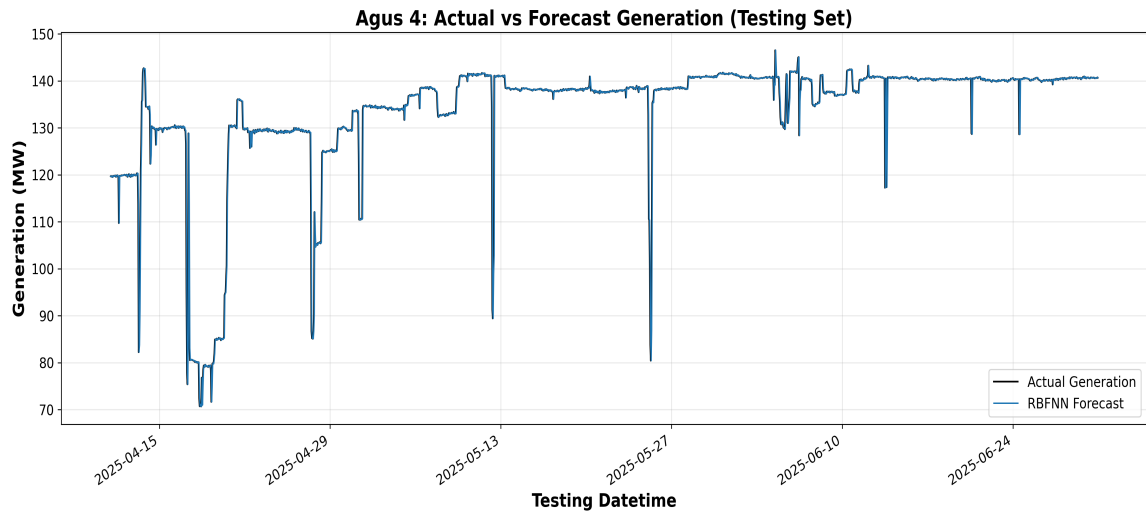


Figure 4.15. Actual and forecasted generation of Agus 4 during the testing period.

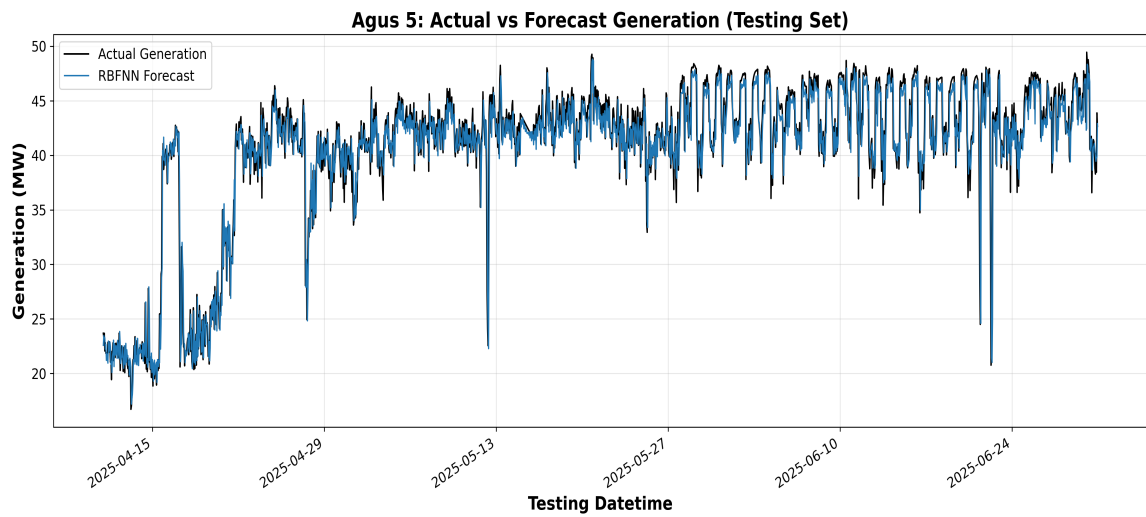


Figure 4.16. Actual and forecasted generation of Agus 5 during the testing period.

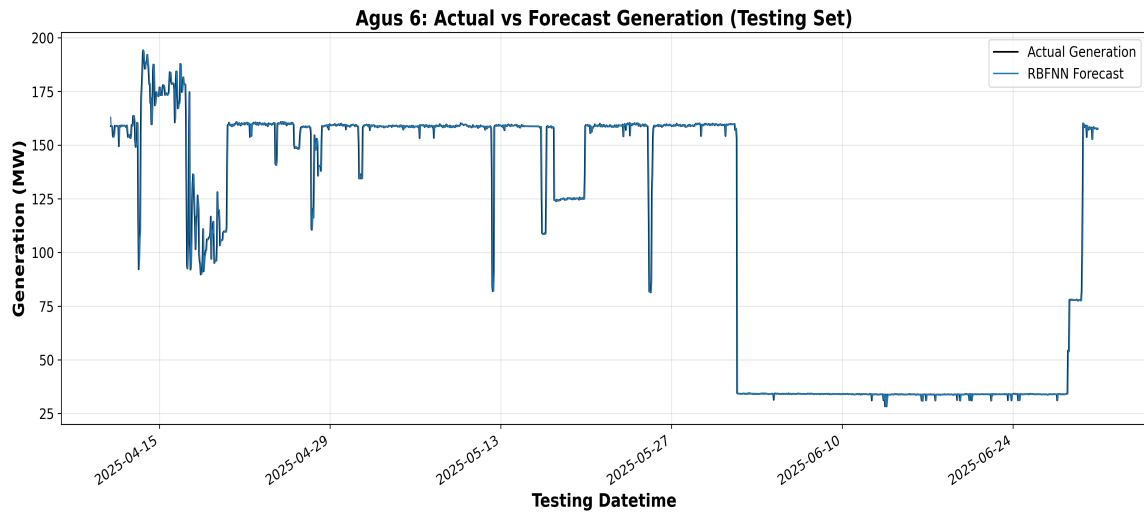


Figure 4.17. Actual and forecasted generation of Agus 6 during the testing period.

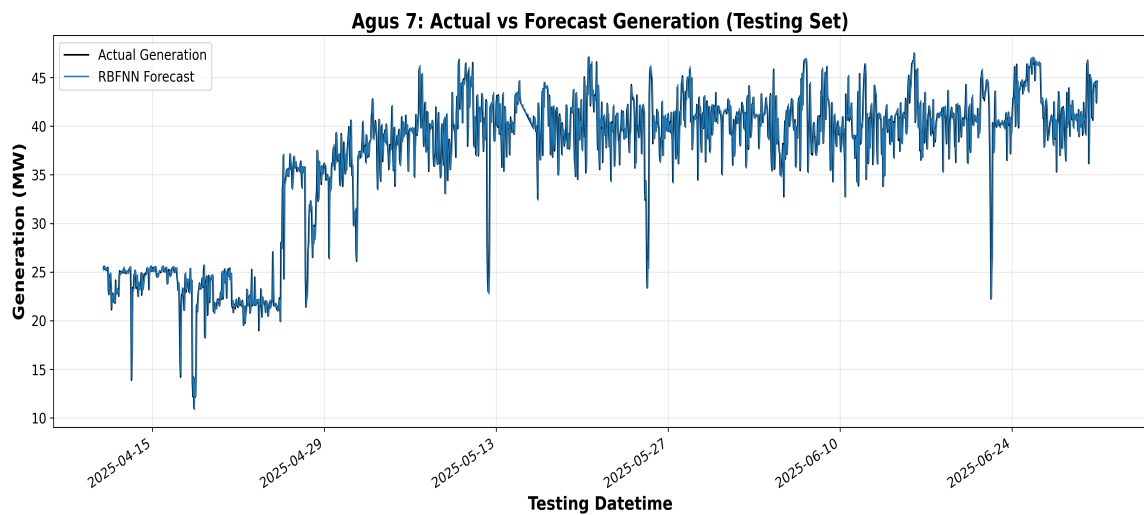


Figure 4.18. Actual and forecasted generation of Agus 7 during the testing period.

Figures 4.13 to 4.18 compare the actual and forecasted generation values of the RBFNN model during the testing period. The testing period represents unseen chronological data from 2025-04-11 to 2025-06-30.

The plots show that the RBFNN forecasts closely follow the actual generation patterns across all six Agus plants, capturing both the timing and magnitude of generation changes. The strongest agreement was observed in Agus 4, while relatively larger deviations were observed in Agus 7.

These observations confirm that the RBFNN model not only achieves strong numer-

ical accuracy but also preserves the temporal structure of hydropower generation, which is critical for operational forecasting.

4.4.2. Testing Error Analysis

To further evaluate the performance of the RBFNN model, additional analyses were conducted using scatter plots, residual distributions, and plant-level error comparisons.

Figure 4.19 shows the relationship between actual and predicted generation values for all plants during the testing period. Figure 4.20 presents the distribution of forecast residuals, while Figure 4.21 compares the testing MAPE across the six Agus plants.

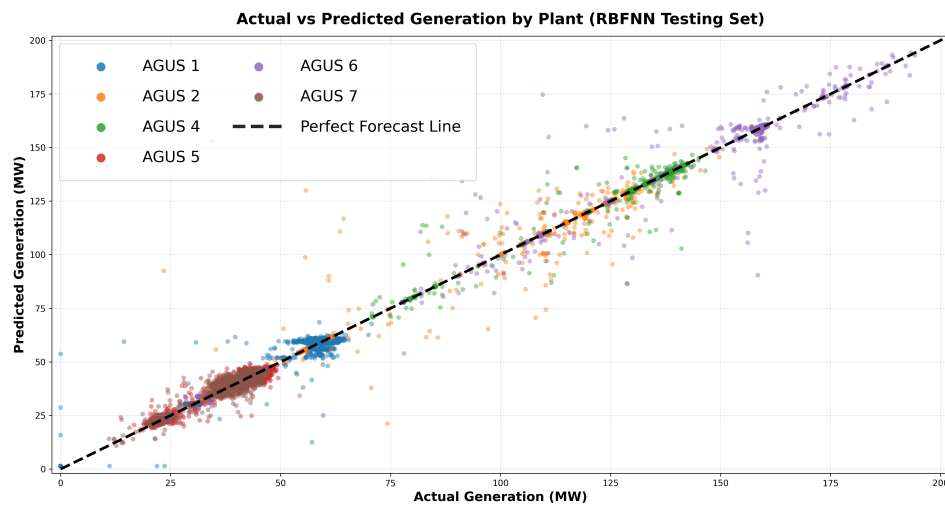


Figure 4.19. Scatter plot of actual and predicted generation values during testing.

The scatter plot shows that most predicted values are closely aligned with the ideal forecast line, indicating strong agreement between actual and predicted generation and confirming the high correlation observed in the testing metrics.

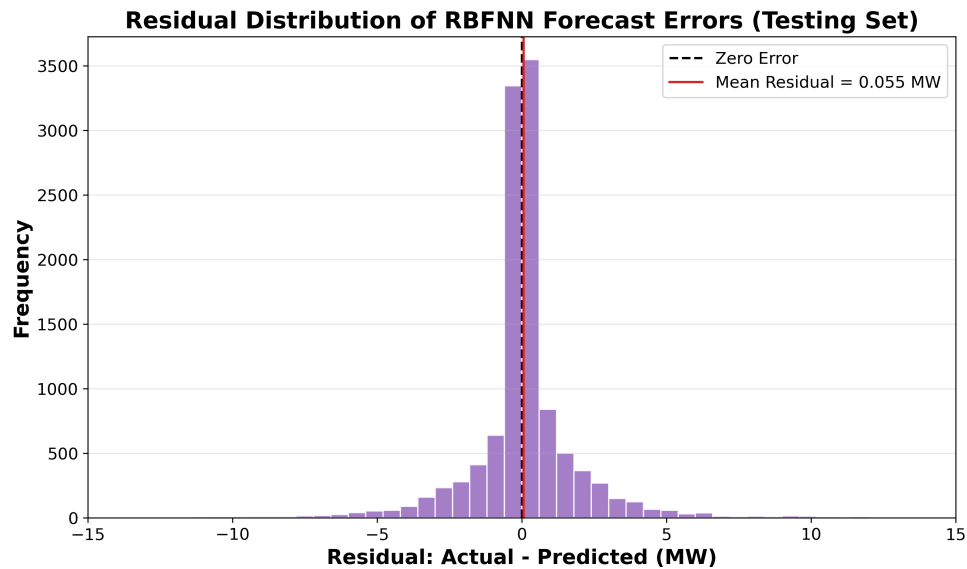


Figure 4.20. Residual distribution of RBFNN forecast errors during testing.

The residual distribution is centered near zero, with a mean residual of 0.055 MW and a median residual of 0.023 MW, indicating that the model does not exhibit strong systematic overprediction or underprediction. This suggests that the forecasting errors are stable and largely unbiased.

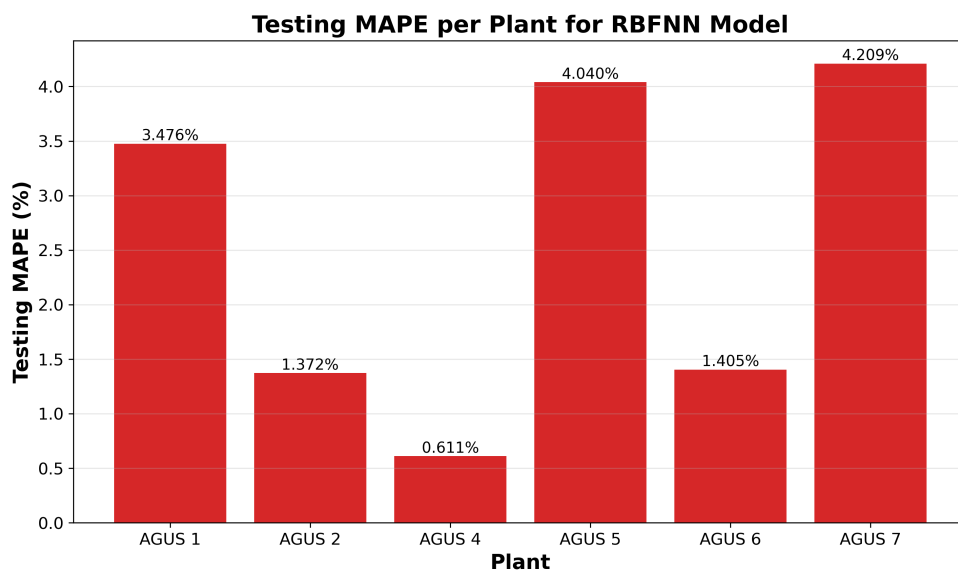


Figure 4.21. Testing MAPE per Agus plant using the RBFNN model.

The MAPE comparison shows that Agus 4 achieved the lowest testing MAPE at 0.611%, while Agus 7 recorded the highest testing MAPE at 4.209%. Overall, all plants

remained below 10% MAPE, further supporting the reliability of the RBFNN model for short-term generation forecasting.

Taken together, these results demonstrate that the RBFNN model provides accurate, stable, and unbiased forecasts, reinforcing its suitability for real-world hydropower generation prediction.

4.5. Benchmark Model Comparison

This section presents a comparative evaluation of the proposed RBFNN model against two benchmark models, namely Random Forest (RF) and XGBoost. The comparison is conducted using the testing dataset to ensure an unbiased assessment of model generalization performance.

To provide a structured evaluation, the comparison is divided into three components: quantitative performance assessment, error analysis, and sample forecast comparison.

4.5.1. Quantitative Performance Comparison

The evaluation focuses on standard forecasting metrics, including Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). These metrics provide a comprehensive assessment of both accuracy and variability in model predictions.

Tables 4.8, 4.9, and 4.10 summarize the quantitative performance of the benchmark models and the proposed RBFNN model.

Table 4.8. Random Forest Testing Performance per Agus Plant

Plant	MAPE (%)	MAE (MW)	RMSE (MW)	R^2
Agus 1	3.486	1.875	3.428	0.900
Agus 2	2.161	1.934	5.005	0.905
Agus 4	1.164	1.365	3.454	0.931
Agus 5	4.200	1.598	2.269	0.893
Agus 6	19.879	8.033	12.198	0.956
Agus 7	4.294	1.527	2.175	0.911

Table 4.9. XGBoost Testing Performance per Agus Plant

Plant	MAPE (%)	MAE (MW)	RMSE (MW)	R^2
Agus 1	3.573	1.928	3.508	0.895
Agus 2	2.554	2.173	5.219	0.896
Agus 4	1.051	1.266	3.273	0.938
Agus 5	4.228	1.614	2.233	0.897
Agus 6	37.541	14.058	22.579	0.850
Agus 7	4.311	1.542	2.174	0.911

Table 4.10. Overall Testing Performance Comparison of Forecasting Models

Model	MAPE (%)	MAE (MW)	RMSE (MW)	R^2
RBFNN	2.519	1.355	3.575	0.925
Random Forest	5.864	2.722	4.755	0.916
XGBoost	8.876	3.764	6.498	0.898

The results demonstrate that the RBFNN model consistently outperformed the benchmark models across most evaluation metrics. As shown in Table 4.10, the RBFNN

achieved the lowest overall testing MAPE of 2.519%, compared to 5.864% for Random Forest and 8.876% for XGBoost. Similarly, the RBFNN yielded lower MAE and RMSE values, indicating more accurate and stable predictions.

At the individual plant level, the superiority of the RBFNN model is particularly evident in complex operating conditions. For instance, Agus 6 exhibited significantly higher errors for the benchmark models, with MAPE values reaching 19.879% for Random Forest and 37.541% for XGBoost, while the RBFNN maintained a relatively low error of 1.405%. This suggests that tree-based models struggled to generalize effectively under highly dynamic operating conditions, whereas the RBFNN was able to better capture nonlinear temporal dependencies.

4.5.2. Error Analysis and Visualization

To further analyze model performance beyond numerical metrics, visual comparisons are presented. Figures 4.22 and 4.23 illustrate the comparative error distributions across all Agus plants.

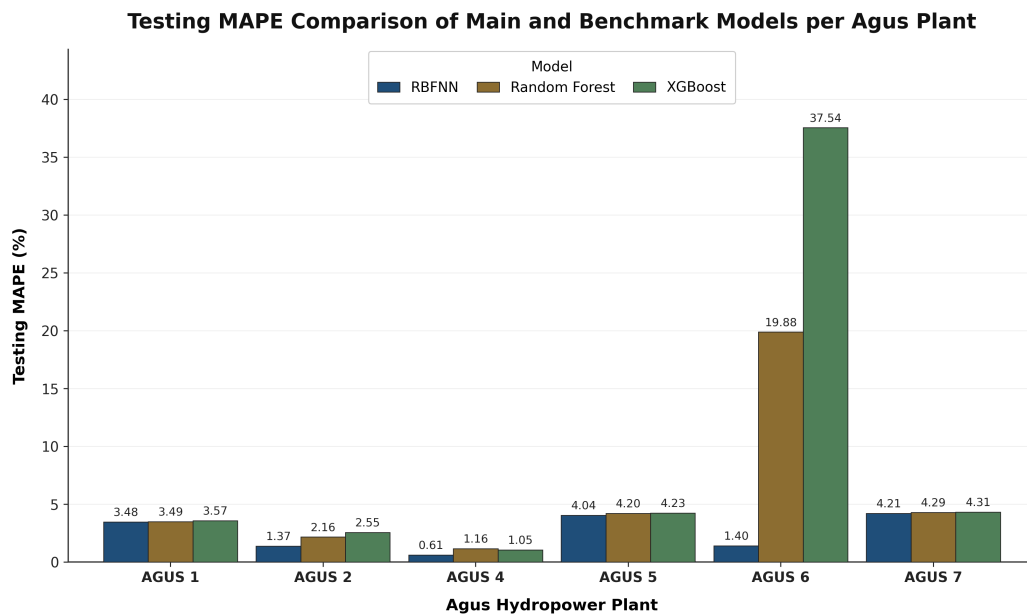


Figure 4.22. Comparison of testing MAPE across Agus plants for RBFNN, Random Forest, and XGBoost models.

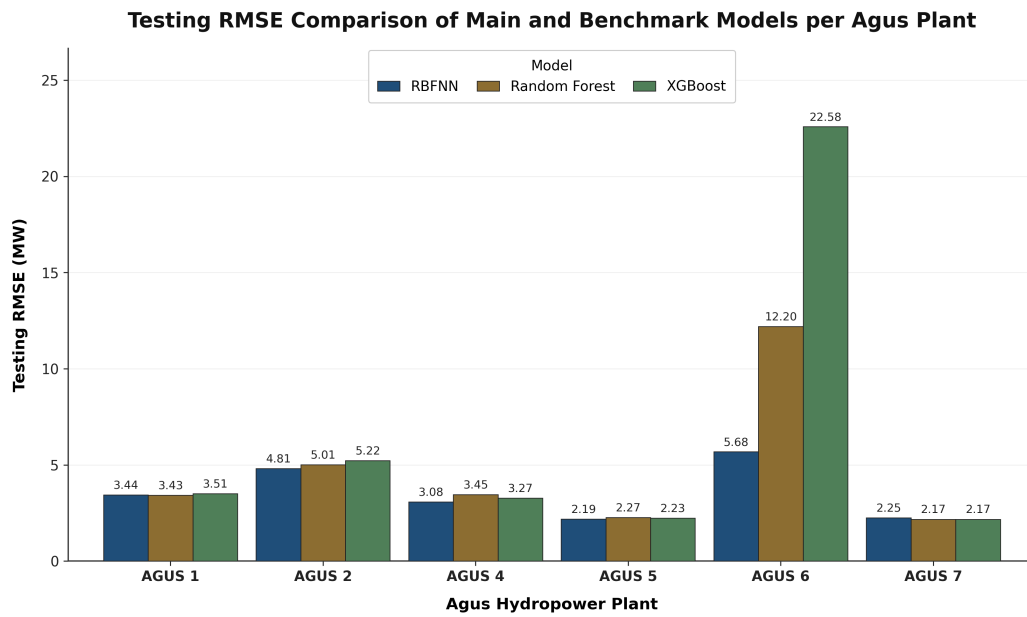


Figure 4.23. Comparison of testing RMSE across Agus plants for RBFNN, Random Forest, and XGBoost models.

The comparison plots in Figures 4.22 and 4.23 further confirm the consistent performance of the RBFNN model across all plants. The benchmark models show greater variability and sensitivity, particularly in plants with more complex hydropower dynamics. This behavior suggests that tree-based models are less effective in capturing temporal dependencies in hydropower systems, particularly under fluctuating operational conditions.

4.5.3. Sample Forecast Comparison

Furthermore, Figures 4.24–4.29 present sample forecast comparisons for a representative testing day.

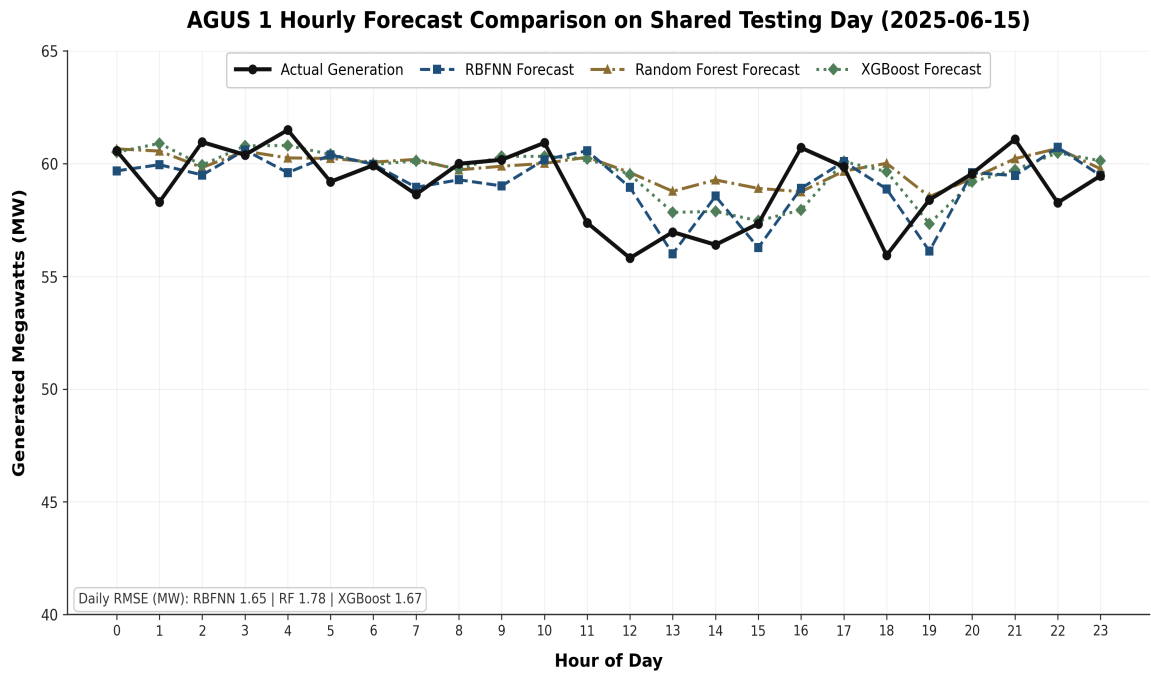


Figure 4.24. Actual vs forecast generation for Agus 1 using the RBFNN model.

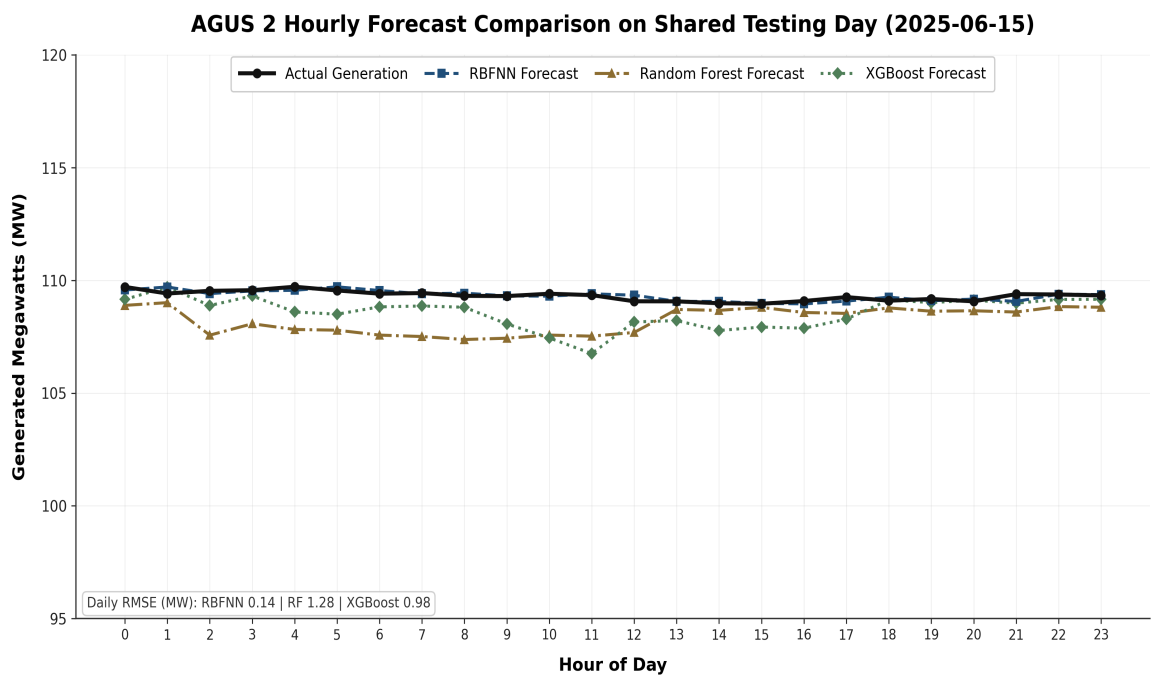


Figure 4.25. Actual vs forecast generation for Agus 2 using the RBFNN model.

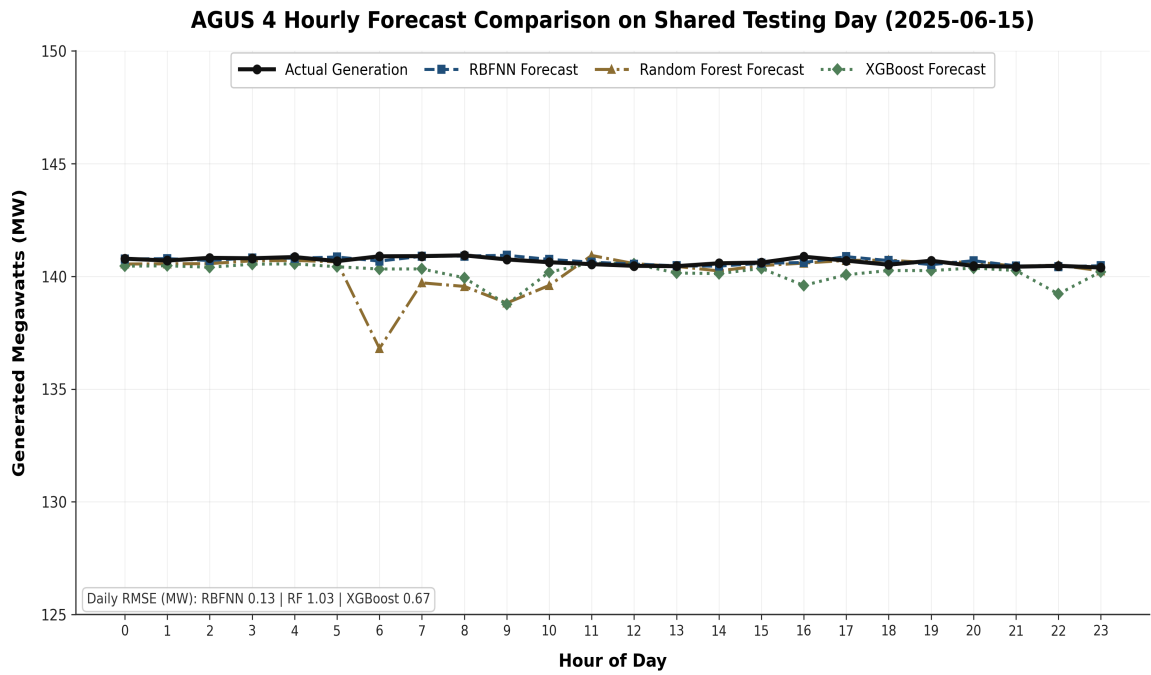


Figure 4.26. Actual vs forecast generation for Agus 4 using the RBFNN model.

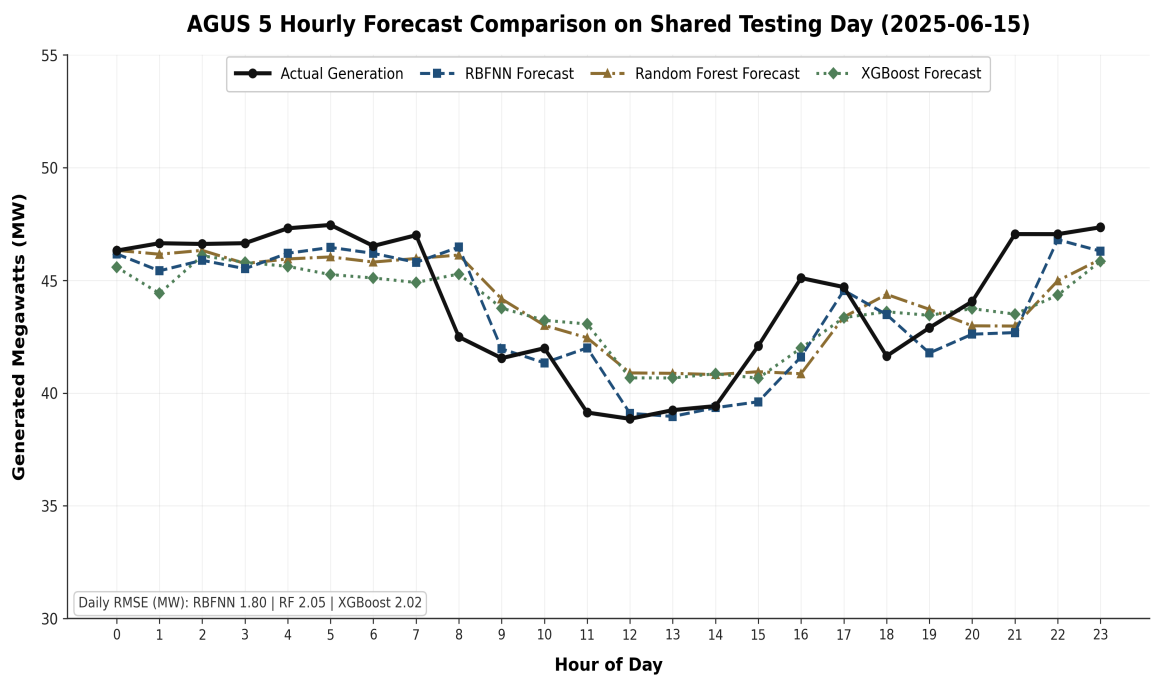


Figure 4.27. Actual vs forecast generation for Agus 5 using the RBFNN model.

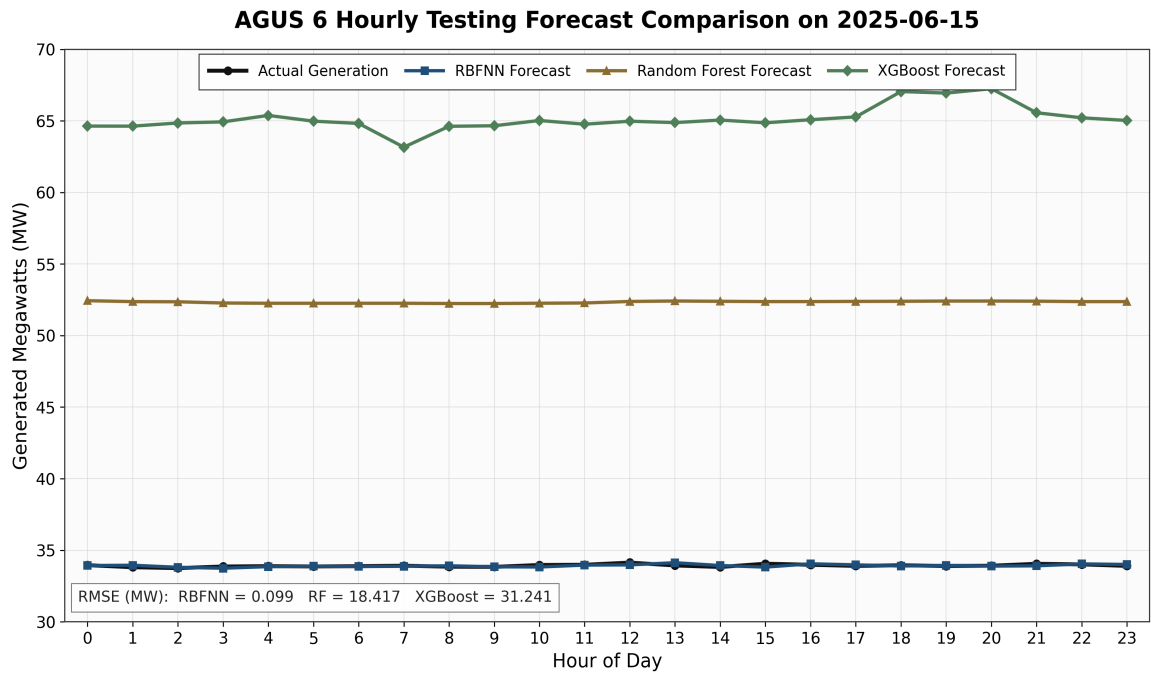


Figure 4.28. Actual vs forecast generation for Agus 6 using the RBFNN model.

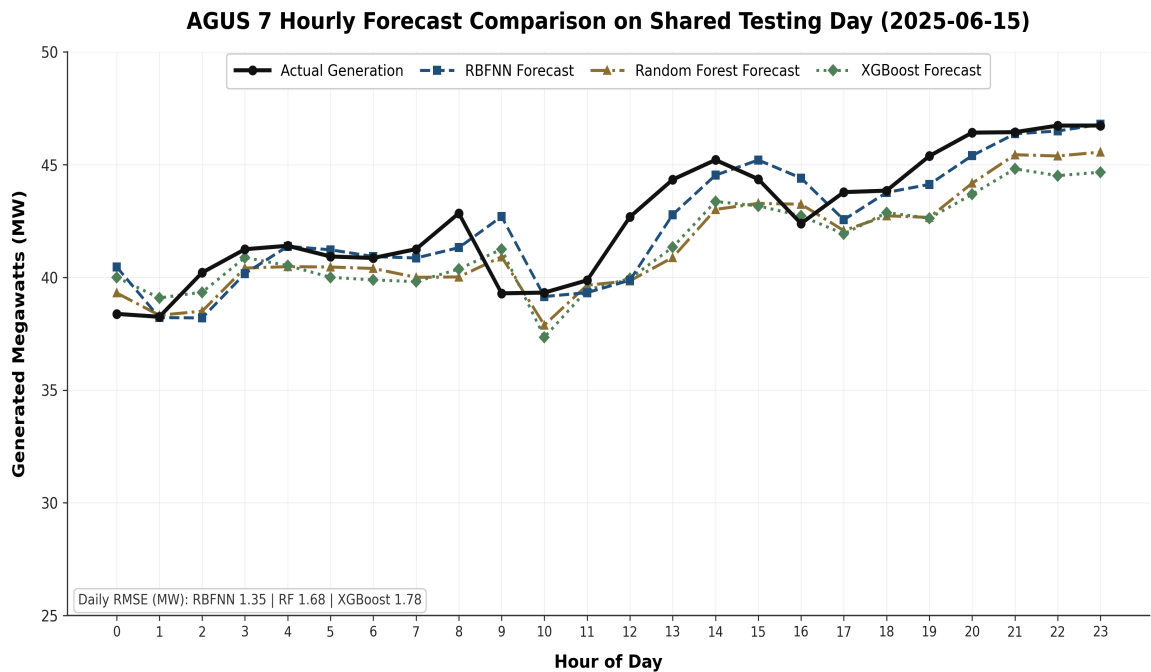


Figure 4.29. Actual vs forecast generation for Agus 7 using the RBFNN model.

The sample forecast comparisons in Figures 4.24–4.29 provide additional insight into model behavior. The RBFNN predictions closely follow the actual generation patterns, demonstrating strong temporal tracking and consistent alignment with observed val-

ues across hourly intervals. This indicates that the model effectively captures both short-term fluctuations and underlying generation trends. In contrast, the Random Forest and XGBoost models tend to deviate during periods of rapid change, indicating weaker temporal modeling capability.

Overall, the results confirm that the RBFNN model provides a more robust and reliable approach for short-term hydropower generation forecasting. Its superior performance across multiple evaluation metrics, combined with its ability to capture nonlinear relationships and temporal dependencies, leads to improved predictive accuracy and stability.

Compared to tree-based models, the RBFNN demonstrates stronger generalization capability, particularly under complex and dynamic operating conditions, making it more suitable for real-world deployment in cascaded hydropower systems.

These results further highlight the advantage of the RBFNN model in handling nonlinear and time-dependent characteristics of hydropower generation.

4.6. Day-Ahead Forecast

This section presents the final operational output of the developed forecasting framework using the selected RBFNN model, which is adopted as the deployable solution for day-ahead forecasting. The system generates hourly forecasts for the Agus cascade based on the most recent available data while incorporating planned outage schedules at the unit level. Unlike the previous sections, which focused on model evaluation and benchmarking, this section demonstrates the practical deployment of the forecasting system.

To illustrate the operational workflow, the day-ahead forecasting process is presented in terms of input configuration, model output, and system-level behavior.

4.6.1. Forecast Input: Planned Outage Schedule

The forecasting process incorporates a planned outage schedule to reflect real-world operational constraints. The outage input is defined at the unit level, where a value of 1 indicates an available generating unit and 0 represents an outage condition.

Planned Outages Input for Forecasting

Date	Hour	out agus1 unit1	out agus1 unit2	out agus2 unit1	out agus2 unit2	out agus2 unit3	out agus4 unit1	out agus4 unit2	out agus4 unit3	out agus5 unit1	out agus5 unit2	out agus6 unit1	out agus6 unit2	out agus6 unit3	out agus6 unit4	out agus6 unit5	out agus7 unit1	out agus7 unit2
2025-07-01	00:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	1:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	2:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	3:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	4:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	5:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	6:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	7:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	8:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	9:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	10:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	11:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	12:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	13:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	14:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	15:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	16:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	17:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	18:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	19:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	20:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	21:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	22:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00
2025-07-01	23:00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0	1.00	1.00	1.00	1.00	1.00

Figure 4.30. Planned outage schedule used as input for day-ahead forecasting.

As shown in Figure 4.30, specific generating units are scheduled for outage during the forecast horizon. These outage conditions directly influence the predicted generation output by constraining the available capacity of each plant.

4.6.2. Day-Ahead Forecast Output

The RBFNN model generates hourly forecasts for each unit and plant in the Agus cascade for the next 24-hour (day-ahead) horizon. The output includes unit-level generation, plant-level totals, and overall cascade generation.

Due to the large size of the full forecast table, the results are presented in their original operational format as shown in Figure 4.31.

RBFNN Day-Ahead 24H Forecast Final Output

Date	Hour	Gen Agus1 U1 MW	Gen Agus1 U2 MW	Gen Agus2 U1 MW	Gen Agus2 U2 MW	Gen Agus2 U3 MW	Gen Agus4 U1 MW	Gen Agus4 U2 MW	Gen Agus4 U3 MW	Gen Agus5 U1 MW	Gen Agus5 U2 MW	Gen Agus5 U3 MW	Gen Agus6 U1 MW	Gen Agus6 U2 MW	Gen Agus6 U3 MW	Gen Agus6 U4 MW	Gen Agus6 U5 MW	Gen Agus7 U1 MW	Gen Agus7 U2 MW	Total Agus1 MW	Total Agus2 MW	Total Agus4 MW	Total Agus5 MW	Total Agus6 MW	Total Agus7 MW	Total Cascade MW
2025-07-0	00:00	30.09	30.09	43.32	43.32	43.32	46.89	46.89	46.89	21.43	21.43	29.46	0.00	42.69	42.69	42.69	22.12	22.12	60.17	129.97	140.66	42.86	157.53	44.24	575.43	
2025-07-0	1:00	30.09	30.09	43.32	43.32	43.32	46.89	46.89	46.89	23.62	23.62	29.45	0.00	42.69	42.69	42.69	22.25	22.25	60.17	129.96	140.66	47.24	157.52	44.51	580.07	
2025-07-0	2:00	30.08	30.08	43.32	43.32	43.32	46.89	46.89	46.89	24.01	24.01	29.45	0.00	42.69	42.69	42.69	22.44	22.44	60.17	129.96	140.66	48.01	157.51	44.88	581.20	
2025-07-0	3:00	30.08	30.08	43.32	43.32	43.32	46.89	46.89	46.89	23.47	23.47	29.45	0.00	42.68	42.68	42.68	22.61	22.61	60.17	129.96	140.67	46.93	157.51	45.22	580.45	
2025-07-0	4:00	30.08	30.08	43.32	43.32	43.32	46.89	46.89	46.89	24.04	24.04	29.45	0.00	42.68	42.68	42.68	22.54	22.54	60.17	129.95	140.67	48.08	157.50	45.07	581.44	
2025-07-0	5:00	30.08	30.08	43.32	43.32	43.32	46.89	46.89	46.89	23.88	23.88	29.45	0.00	42.68	42.68	42.68	21.57	21.57	60.17	129.95	140.67	47.77	157.49	43.15	579.20	
2025-07-0	6:00	30.08	30.08	43.32	43.32	43.32	46.89	46.89	46.89	23.52	23.52	29.45	0.00	42.68	42.68	42.68	21.20	21.20	60.17	129.95	140.67	47.04	157.49	42.39	577.70	
2025-07-0	7:00	30.08	30.08	43.31	43.31	43.31	46.89	46.89	46.89	21.14	21.14	29.45	0.00	42.68	42.68	42.68	21.48	21.48	60.17	129.94	140.67	42.28	157.48	42.97	573.51	
2025-07-0	8:00	30.08	30.08	43.31	43.31	43.31	46.89	46.89	46.89	20.57	20.57	29.45	0.00	42.68	42.68	42.68	21.73	21.73	60.17	129.94	140.67	41.14	157.47	43.45	572.85	
2025-07-0	9:00	30.08	30.08	43.31	43.31	43.31	46.89	46.89	46.89	20.57	20.57	29.45	0.00	42.67	42.67	42.67	21.82	21.82	60.17	129.93	140.68	41.13	157.47	43.63	573.01	
2025-07-0	10:00	30.08	30.08	43.31	43.31	43.31	46.89	46.89	46.89	20.81	20.81	29.44	0.00	42.67	42.67	42.67	21.85	21.85	60.17	129.93	140.68	41.63	157.46	43.69	573.56	
2025-07-0	11:00	30.08	30.08	43.31	43.31	43.31	46.89	46.89	46.89	19.05	19.05	29.44	0.00	42.67	42.67	42.67	21.56	21.56	60.17	129.93	140.68	38.10	157.45	43.13	569.45	
2025-07-0	12:00	30.08	30.08	43.31	43.31	43.31	46.89	46.89	46.89	19.59	19.59	29.44	0.00	42.67	42.67	42.67	21.56	21.56	60.17	129.92	140.68	39.18	157.45	43.13	570.52	
2025-07-0	13:00	30.08	30.08	43.31	43.31	43.31	46.89	46.89	46.89	20.14	20.14	29.44	0.00	42.67	42.67	42.67	21.35	21.35	60.17	129.92	140.68	40.29	157.44	42.71	571.20	
2025-07-0	14:00	30.08	30.08	43.31	43.31	43.31	46.89	46.89	46.89	20.56	20.56	29.44	0.00	42.66	42.66	42.66	21.50	21.50	60.17	129.92	140.68	41.12	157.43	43.00	572.32	
2025-07-0	15:00	30.08	30.08	43.30	43.30	43.30	46.90	46.90	46.90	20.62	20.62	29.44	0.00	42.66	42.66	42.66	21.64	21.64	60.16	129.91	140.69	41.24	157.43	43.27	572.70	
2025-07-0	16:00	30.08	30.08	43.30	43.30	43.30	46.90	46.90	46.90	20.34	20.34	29.44	0.00	42.66	42.66	42.66	21.74	21.74	60.16	129.91	140.69	40.67	157.42	43.49	572.34	
2025-07-0	17:00	30.08	30.08	43.30	43.30	43.30	46.90	46.90	46.90	19.72	19.72	29.44	0.00	42.66	42.66	42.66	21.87	21.87	60.16	129.90	140.69	39.44	157.41	43.74	571.36	
2025-07-0	18:00	30.08	30.08	43.30	43.30	43.30	46.90	46.90	46.90	19.32	19.32	29.43	0.00	42.66	42.66	42.66	21.92	21.92	60.16	129.90	140.69	38.63	157.41	43.85	570.64	
2025-07-0	19:00	30.08	30.08	43.30	43.30	43.30	46.90	46.90	46.90	20.02	20.02	29.43	0.00	42.66	42.66	42.66	22.11	22.11	60.16	129.90	140.69	40.03	157.40	44.22	572.41	
2025-07-0	20:00	30.08	30.08	43.30	43.30	43.30	46.90	46.90	46.90	19.52	19.52	29.43	0.00	42.65	42.65	42.65	21.94	21.94	60.16	129.89	140.69	39.04	157.39	43.87	571.05	
2025-07-0	21:00	30.08	30.08	43.30	43.30	43.30	46.90	46.90	46.90	21.16	21.16	29.43	0.00	42.65	42.65	42.65	21.99	21.99	60.16	129.89	140.70	42.32	157.39	43.98	574.44	
2025-07-0	22:00	30.08	30.08	43.30	43.30	43.30	46.90	46.90	46.90	21.31	21.31	29.43	0.00	42.65	42.65	42.65	22.07	22.07	60.16	129.89	140.70	42.61	157.38	44.15	574.89	
2025-07-0	23:00	30.08	30.08	43.29	43.29	43.29	46.90	46.90	46.90	21.30	21.30	29.43	0.00	42.65	42.65	42.65	22.10	22.10	60.16	129.88	140.70	42.61	157.37	44.21	574.94	

Figure 4.31. Day-ahead forecast output generated by the RBFNN model in Excel format.

The forecast output follows a format consistent with operational reporting practices in hydropower scheduling systems. It can be observed that when a unit is scheduled for outage, the corresponding generation output is reduced or set to zero at the affected units, which is reflected in both plant-level and overall cascade outputs. This demonstrates that the model successfully integrates operational constraints into the forecasting process.

4.6.3. Total Cascade Forecast

To evaluate the system-level behavior, the total cascade generation forecast is presented in Figure 4.32.

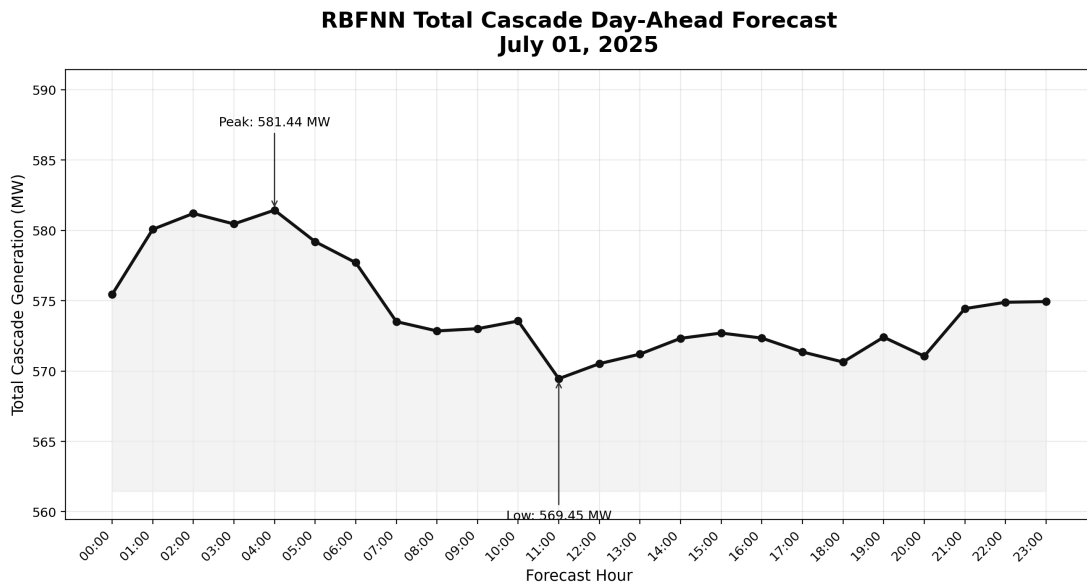


Figure 4.32. Total cascade day-ahead generation forecast using the RBFNN model for July 01, 2025.

The forecast exhibits smooth temporal variation in total generation, with a peak value of approximately 581.44 MW and a minimum value of approximately 569.45 MW. This behavior indicates stable system operation and reflects realistic generation dynamics of the Agus cascade. It also confirms that the model maintains continuity without introducing unrealistic spikes, which is critical for operational reliability.

The integration of planned outage schedules ensures that the forecast remains operationally consistent, as units under outage conditions are appropriately excluded from the predicted output. This confirms the capability of the model to incorporate real-world constraints into the forecasting process.

Overall, the results confirm that the developed RBFNN-based forecasting framework is suitable for practical deployment, providing reliable, constraint-aware day-ahead forecasts for short-term scheduling and capacity nomination.

4.7. Discussion of Results

This section synthesizes the key findings of the study and interprets the results in relation to the research objectives. The discussion focuses on model performance, compari-

son with benchmark approaches, and the practical implications of the developed forecasting framework for hydropower operations.

4.7.1. Model Learning Behavior and Generalization

The training and validation results indicate that the RBFNN models achieved stable and consistent learning behavior across all Agus plants. The relatively flat loss curves observed during training suggest rapid convergence, which is consistent with the use of standardized inputs and the prediction of residual (Δ) generation values.

Despite the minimal variation in loss curves, the validation metrics demonstrate strong predictive performance, with low MAPE values and high coefficients of determination across all plants. This confirms that the model effectively captured the nonlinear relationships inherent in hydropower generation and generalized well to unseen data.

The ability of the RBFNN to learn plant-specific characteristics while maintaining overall stability highlights its suitability for modeling complex and interconnected hydropower systems.

4.7.2. Forecasting Accuracy and Reliability

The testing results further validate the effectiveness of the RBFNN model in short-term generation forecasting. The model achieved an overall testing MAPE of 2.519%, indicating high forecasting accuracy across the Agus cascade.

At the individual plant level, variations in forecasting accuracy were observed. Plants such as Agus 4 exhibited very low error values, reflecting stable and predictable generation patterns. In contrast, downstream plants such as Agus 7 showed relatively higher errors, which can be attributed to increased operational variability and sensitivity to upstream conditions.

The strong alignment between actual and forecasted generation profiles confirms that the model not only performs well numerically but also preserves the temporal dynamics of the system. This is particularly important for operational forecasting, where both accuracy and trend consistency are required.

4.7.3. Comparison with Benchmark Models

The comparative analysis demonstrates that the RBFNN model consistently outperforms the Random Forest and XGBoost models across all evaluation metrics. The superior performance of the RBFNN is particularly evident under complex operating conditions, such as those observed in Agus 6, where the benchmark models exhibited significantly higher errors.

This performance advantage can be attributed to the ability of the RBFNN to capture nonlinear temporal dependencies more effectively than tree-based models. While Random Forest and XGBoost rely on partition-based learning, they are less capable of modeling continuous temporal dynamics, especially in systems with strong interdependencies such as cascaded hydropower plants.

The results highlight the importance of selecting models that are well-suited to the physical and temporal characteristics of the system being studied.

4.7.4. Operational Implications of the Forecasting Framework

Beyond numerical accuracy, the results demonstrate the practical applicability of the developed forecasting framework. The integration of planned outage schedules allows the model to produce constraint-aware forecasts, ensuring that unavailable units do not contribute to the predicted generation output.

At the system level, the total cascade forecast exhibits smooth and continuous behavior, indicating stable operation and realistic representation of generation dynamics. This capability is critical for real-world applications such as capacity nomination, scheduling, and short-term operational planning.

Furthermore, the use of an output format consistent with operational reporting practices enhances the usability of the forecasting system. This ensures that the model outputs can be directly interpreted and utilized by system operators without requiring additional transformation or processing.

4.7.5. Overall Assessment

Overall, the results confirm that the proposed RBFNN-based forecasting approach provides a robust and reliable solution for short-term hydropower generation forecasting. The model demonstrates strong predictive accuracy, stable generalization, and superior performance compared to benchmark methods.

More importantly, the developed framework successfully bridges the gap between model performance and practical deployment. By incorporating operational constraints and preserving system-level behavior, the forecasting system is well-suited for real-world implementation in cascaded hydropower systems such as the Agus cascade.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1. Conclusion

This study developed and evaluated a Radial Basis Function Neural Network (RBFNN)-based framework for short-term generation forecasting of the Agus 1–7 cascaded hydropower system. The primary objective was to design a model capable of accurately predicting day-ahead hourly generation while accounting for the nonlinear and interdependent characteristics of cascaded hydropower operations.

The results demonstrate that the proposed RBFNN model achieved strong predictive performance across all Agus plants. The model consistently produced low forecasting errors, with an overall testing Mean Absolute Percentage Error (MAPE) of 2.519% and high coefficients of determination, indicating strong agreement between predicted and actual generation values. These findings confirm that the RBFNN effectively captured the temporal dynamics and nonlinear behavior inherent in hydropower generation.

In comparison with benchmark models, namely Random Forest and XGBoost, the RBFNN showed superior performance across all evaluation metrics. The model was particularly effective under complex and highly dynamic operating conditions, where the benchmark models exhibited significantly higher errors. This highlights the advantage of the RBFNN in modeling time-dependent and nonlinear systems such as cascaded hydropower plants.

Beyond numerical accuracy, the developed forecasting framework demonstrated practical applicability. The integration of planned outage schedules enabled the model to generate constraint-aware forecasts, ensuring that unavailable generating units were properly excluded from predicted outputs. Additionally, the system-level cascade forecasts exhibited smooth and realistic generation behavior, confirming the model's ability to maintain continuity and operational consistency.

Overall, the results confirm that the proposed RBFNN-based approach provides a reliable and deployable solution for short-term hydropower generation forecasting. The

framework not only improves prediction accuracy but also supports real-world operational requirements such as scheduling, capacity nomination, and decision-making in cascaded hydropower systems.

5.2. Recommendations

Based on the findings of this study, the following recommendations are proposed to further enhance the forecasting framework and extend its applicability:

1. Future studies may explore the integration of additional hydrological and environmental variables, such as rainfall intensity, inflow measurements, and weather forecasts, to further improve model accuracy under varying hydrological conditions.
2. The incorporation of more advanced temporal models, such as Long Short-Term Memory (LSTM) networks or hybrid architectures, may be investigated to enhance the modeling of long-term dependencies and complex temporal patterns in hydropower systems.
3. The forecasting framework may be extended to support multi-step or probabilistic forecasting, allowing operators to quantify uncertainty and improve decision-making under varying operating conditions.
4. The model can be integrated into a real-time decision support system, enabling continuous updates of forecasts based on incoming operational data and improving its practical utility for system operators.
5. It is recommended to develop an automated data acquisition platform that directly collects and updates operational data from plant systems. This would eliminate the need for manual data entry, reduce the risk of human error, and improve the reliability and timeliness of the forecasting process.
6. Future work may include the development of a graphical user interface (GUI) to improve user interaction with the forecasting system. Such an interface would allow

operators to input planned outage schedules and visualize forecast outputs in a more accessible and user-friendly manner.

7. The proposed approach may be applied to other cascaded hydropower systems to evaluate its generalizability and robustness across different operational environments.
8. Further research may explore the integration of forecasting with optimization techniques, enabling the development of a comprehensive system for generation scheduling, water resource management, and operational planning.

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